A Product Manifold Method for Feature Selection

Mao Li
School of Artificial Intelligence, Beihang University
Key Laboratory of Mathematics, Informatics and Behavioral Semantics, Ministry of Education
Beijing, China
limao@buaa.edu.cn

Zhilong Mi*
School of Artificial Intelligence, Beihang University
Beijing Advanced Innovation Center for Future Blockchain and Privacy Computing, Beihang University
Beijing, China
mizhilong9@buaa.edu.cn

Yingpeng Du
Nanyang Technological University
Singapore
yingpeng.du@ntu.edu.sg

Yifan Cao
School of Artificial Intelligence, Beihang University
Key Laboratory of Mathematics, Informatics and Behavioral Semantics, Ministry of Education
Beijing, China
cao_yifan@buaa.edu.cn

Qingcai He
Beijing International Center for Mathematical Research
Beijing, China
Chongqing Big Data Research Institute, Peking University
Chongqing, China
heqingcai@pku.edu.cn

Ziqiao Yin
School of Artificial Intelligence, Beihang University
Beijing Advanced Innovation Center for Future Blockchain and Privacy Computing, Beihang University
Beijing, China
yinziqiao@buaa.edu.cn

Binghui Guo*
School of Artificial Intelligence, Beihang University
Beijing Advanced Innovation Center for Future Blockchain and Privacy Computing, Beihang University
Beijing, China
guobinghui@buaa.edu.cn

Zhu Sun
Singapore University of Technology and Design
Singapore
zhu_sun@sutd.edu.sg

Abstract

Feature selection is indispensable for mitigating overfitting and reducing feature redundancy in high-dimensional scenarios. However, most existing approaches rely on unstable overall separability and sample-wise geometry, thereby neglecting the stable class-specific discriminative structures and leading to poor performance especially in small sample regimes. Although product manifolds provide a natural geometric framework to model distinctive manifolds for different classes, their potential in feature selection is largely unexplored, e.g., theoretical guarantee on spectral convergence. In this paper, we propose a novel supervised feature selection method named PRISM, which leverages product manifold to theoretically disentangle class-specific features from the shared structure. Grounded in spectral analysis on product manifolds, we explicitly model the feature space to distinguish between shared and class-specific structures. To disentangle these class-specific structures, we design a spectral filtering mechanism that suppresses the shared components and iteratively extracts class-specific latent variables. Based on these extracted variables, we establish a scoring mechanism that identifies features with both high discriminative power and strong class specificity in high-dimensional small-sample scenarios. Crucially, we bridge the theoretical gap in spectral analysis and provide a theoretical guarantee for our method by deriving an asymptotic convergence proof under the product manifold setting, guaranteeing the reliable isolation of class-specific discriminative structures. Comprehensive experiments demonstrate that PRISM not only improves generalization performance and robustness to small sample size over leading baselines, but also achieves superior result reusability when new classes emerge.

CCS Concepts

• **Computing methodologies → Machine learning algorithms**.

Keywords

Feature selection, Manifold learning, Spectral analysis

ACM Reference Format:

Mao Li, Zhilong Mi, Yingpeng Du, Yifan Cao, Qingcai He, Ziqiao Yin, Binghui Guo, and Zhu Sun. 2026. A Product Manifold Method for Feature Selection. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD '26), August 09–13, 2026, Jeju Island, Republic of Korea.* ACM, New York, NY, USA, 18 pages. https://doi.org/10.1145/3770855.3817798

Resource Availability:

The source code is available at https://doi.org/10.5281/zenodo.20439488.

*Corresponding authors.

KDD '26, Jeju Island, Republic of Korea

ACM ISBN 979-8-4007-2259-2/2026/08
https://doi.org/10.1145/3770855.3817798

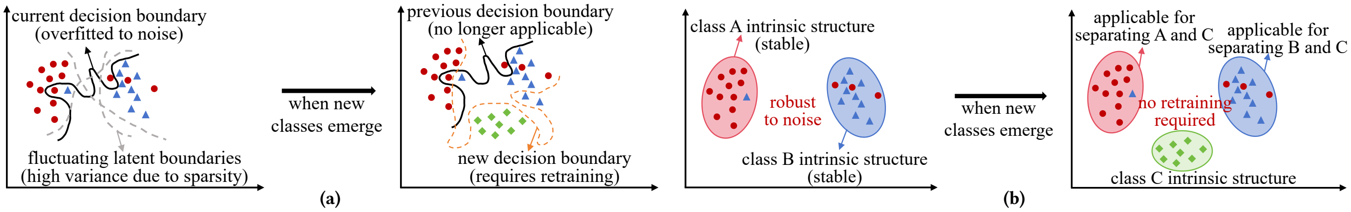

Figure 1: Illustrative comparison: common vs. class-specific discriminative features in small-sample Settings. (a) Existing approaches: identify common features to maximize the overall separability. (b) Proposed approach: identify class-specific features based on the intrinsic structure.

1 Introduction

High-dimensional data is an increasingly common problem faced in the development of models in natural language processing [6, 46] and computer vision [35, 39], which arises directly from a wide range of domains such as medical and clinical research [11, 12, 30], bioinformatics [15, 22], and chemometrics [37]. Traditional models are prone to overfitting, incur prohibitive computational costs, and suffer from severe feature redundancy when feature dimensionality far exceeds sample size [29]. In this context, feature selection plays a critical role in the analysis of high-dimensional datasets. Good feature selection methods can enhance model performance, reduce experimental costs, improve model interpretability, and, more importantly, assist in uncovering the process of data generation [5].

Existing feature selection methods can be classified into three categories: wrapper, embedded, and filter methods. Wrapper methods require model training on different combinations of features, which incurs high computational costs [1, 36]. Embedded methods integrate feature selection into model training, but the selected features depend heavily on the specific model used [9, 43]. Moreover, both methods are prone to overfitting because the feature selection process is guided by the model's performance on training data [40]. In contrast, filter methods rank features independently of any learning model, relying only on statistical or geometric criteria [9], thereby being computationally friendly and robust to overfitting, which makes them well-suited for high-dimensional data.

Despite their advantages, existing filter methods [9, 10, 16, 19–21, 38, 41] suffer from key limitations, especially when the sample size is small. Primarily, current approaches, including both univariate and manifold-based methods, typically rely on simple relevance measures or global sample-wise geometry [10, 16, 19, 31, 48]. Consequently, they overlook the inter-feature interactions, which provide critical robustness against noise in small-sample regimes [9]. Specifically, they predominantly identify common discriminative features shared across classes, failing to uncover class-specific features that uniquely characterize individual classes. Mathematically, these methods generally seek a **single shared mapping operator** f from the input feature space $\mathcal{X}$ to the class label space $\mathcal{Y}$:

$$f : \mathcal{X} \to \mathcal{Y},$$

where f is optimized to identify a shared feature subset (i.e., common features) capable of simultaneously resolving the decision boundaries between all pairs of classes. As illustrated in Figure 1a, relying on this shared operator is particularly precarious in small-sample regimes. Since a single f is constrained to accommodate the complex decision boundaries between every pair of classes simultaneously, the solution becomes inherently unstable and highly susceptible to noise given the sparsity of the data distribution.

In contrast, capturing the intrinsic essence of each category independently provides a theoretically robust alternative, as the manifold of an individual class is inherently stable [16, 48]. Rather than enforcing a single shared feature subset to distinguish all classes, we advocate for identifying **distinct mapping operators** $\{f_k\}_{k=1}^{K}$, each grounded in a unique feature subspace (i.e., class-specific features), to effectively capture intrinsic essence of each category. Each operator f_k selects and leverages class-specific patterns to exclusively identify the k-th class, mapping samples from the input feature space $\mathcal{X}$ to a dedicated binary indicator (where 1 denotes the k-th class, and 0 otherwise) for the k-th class:

$$f_k : \mathcal{X} \to \{0, 1\}.$$

As shown in Figure 1b, by anchoring feature selection on these stable internal structures rather than fluctuating boundaries, class-specific signals offer superior robustness against overfitting. Unlike decision boundaries based on common features, which are highly sensitive to the relative positions of sparse samples from different classes, the intrinsic structure of a class is defined solely by its own data distribution. Consequently, the extracted class-specific features representing this structure remain robust to inter-class noise. Furthermore, such features possess better reusability in evolving small-sample scenarios, as the validity of f_k remains independent of new emerging classes. Therefore, a novel framework is required that not only models feature synergy but, more importantly, leverages the geometric structure of the feature space to uncover class-specific discriminative structures.

To address this challenge in high-dimensional small-sample scenarios, we propose a novel filter-based feature selection method, termed **P**roduct manifold **R**epresentation for **I**solating latent variables via **S**pectral filtering **M**echanism (PRISM). Unlike traditional methods that seek a single subset of common discriminative features, PRISM is designed to extract class-specific latent variables by leveraging the manifold structure of the feature space. Motivated by the decomposition property of product manifolds, which characterizes the data space as the Cartesian product of a shared manifold representing the shared structure and a distinctive manifold capturing the class-specific structure, we adopt this framework to model the feature space and theoretically disentangle class-specific features from the shared structure. To achieve this separation, PRISM implements a *spectral filtering mechanism* that leverages feature association kernels and Laplacian matrices to suppress the shared components and extract class-specific latent variables. By iteratively extracting these class-specific latent variables, we define a feature score that can identify features with high discriminative power and strong class specificity in high-dimensional small-sample scenarios. Crucially, we provide a rigorous theoretical guarantee for our method. While existing spectral analysis theories establish convergence guarantees for single manifold, there is a lack of rigorous theoretical foundation concerning spectral convergence under the product manifold setting. We bridge this gap by deriving an asymptotic convergence proof, ensuring that the extracted latent variables reliably capture the unique intrinsic structure of each class. Specifically, our theoretical analysis establishes a critical guarantee: it proves that the leading eigenvector of our constructed filtered operator asymptotically converges to the eigenfunction of the class-specific manifold, thereby theoretically isolating the class-specific latent variables from the shared structure.

Our contributions are summarized as follows: (a) We propose a novel supervised feature selection method that explicitly models feature synergy and uncovers class-specific discriminative structures via a product manifold framework. By implementing a spectral filtering mechanism to isolate these structures, our approach significantly improves the reliability and robustness of feature selection in high-dimensional small-sample settings. (b) To the best of our knowledge, this is the first work to integrate product manifold with spectral analysis for feature-space manifold learning. We provide an asymptotic convergence analysis for our method, theoretically guaranteeing the feasibility of uncovering class-specific discriminative structures from the shared structure. (c) We conduct comprehensive experiments on benchmark datasets to show that PRISM possesses superior result reusability when new classes are introduced, while consistently outperforming state-of-the-art baselines in terms of generalization capability and robustness to small sample size.

2 Related Work

Existing filter methods can be broadly categorized into three streams based on their evaluation criteria: statistical metric-based, geometry-based, and manifold-based methods.

Statistical and Geometry-based Methods. The first category includes classical methods such as ANOVA F-value [20], Fisher score [10], Information Gain (IG) [41], and Gini-index [38], as well as the recent method PHet [27]. These methods independently evaluate the statistical association between a single feature and the target label. The second category includes geometry-based methods like Laplacian Score [16, 25], SPEC [48], and Relief [21, 31]. These approaches assess feature importance by exploiting the geometric structure of the sample space. Specifically, Laplacian Score and SPEC employ graph Laplacian matrices from sample affinities to capture global relationships, whereas Relief analyzes the geometric configuration of nearest-neighbor samples. However, these methods face significant challenges. By relying on sample-space geometry, they often struggle in high-dimensional small-sample settings where the sample distribution is sparse and noisy. Furthermore, both categories typically prioritize overall separability or univariate associations, thereby failing to uncover class-specific features that uniquely characterize individual classes. **Manifold-based Methods.** The third category involves approaches that leverage the geometric structure of the data space, such as Inf-FS [33], ILFS [32], ManiFeSt [9] and MMDUFS [44]. Inf-FS and ILFS construct feature graphs to model dependencies, utilizing infinite path integration or latent factors to rank features. However, they predominantly yield a global ranking based on centrality, neglecting to disentangle class-specific structures from the shared structure. ManiFeSt relies on the Riemannian manifold of the feature space. However, by constructing a composite kernel, it is limited to capturing the common discriminative features between classes. Consequently, it cannot extract class-specific latent variables that reveal unique intra-class patterns, potentially resulting in the selection of redundant features and limiting reusability when new classes emerge. MMDUFS constructs a shared structure operator and a differential graph operator under the manifold assumption. However, it is designed as a multi-modal, unsupervised differentiable gating framework and employs graph Laplacian matrices based on the sample space to characterize latent structures.

In contrast to these methods, our approach is grounded in the product manifold structure of the feature space. We explicitly model multivariate feature associations to extract class-specific latent variables, thereby enhancing feature reusability and model robustness.

3 Method

We detail the PRISM proposed for feature selection. Our approach is grounded in spectral analysis on product manifolds to extract class-specific latent variables that capture the unique intrinsic structures of different classes. Figure 2 illustrates the overview of PRISM.

3.1 Manifold Assumption

While the feature space is high-dimensional, the data effectively resides on low-dimensional manifolds governed by a small number of latent variables. For two datasets drawn from different classes within the same modality, their latent spaces may follow a similar underlying structure while also exhibiting unique intra-class structural variations. For illustration, consider two sets $X^{(1)}$ and $X^{(2)}$ in Figure 3, each representing a distinct class within a single dataset X. Let the observations $x_i^{(1)} \in X^{(1)}$ and $x_i^{(2)} \in X^{(2)}$. Each observation from the dataset can be approximated by the result of a continuous transformation applied to a set of latent variables,

$$(\theta_i, \varphi_i^{(1)}) \xrightarrow{T_1} x_i^{(1)}, \quad (\theta_i, \varphi_i^{(2)}) \xrightarrow{T_2} x_i^{(2)}. \tag{1}$$

The shared latent variables θ capture the structure shared by both classes $X^{(1)}$ and $X^{(2)}$(the similar tree structure), whereas the class-specific latent variables $\varphi^{(i)}$ capture the structure specific to $X^{(i)}$(the additional branch in the tree of $X^{(2)}$ but absent in $X^{(1)}$).

By extracting these class-specific variables, we can identify features associated with the unique intrinsic structure of each class, thus precisely characterizing the differential patterns and identifying the key discriminative features. To achieve this goal, we introduce a product manifold framework to explicitly model class-specific discriminative structures in the feature space.

3.2 The Framework of Product Manifolds

To formally characterize the separation of shared and class-specific latent variables, we model the feature space via product manifolds. This framework enables the rigorous decomposition of data geometry into independent shared and class-specific structures.

Definition 1 (Product Manifold). *Let $\mathcal{M}_a$ and $\mathcal{M}_b$ be two manifolds. The product manifold $\mathcal{M}$ is defined as the Cartesian product of the underlying sets:*

$$\mathcal{M} := \mathcal{M}_a \times \mathcal{M}_b = \{(p, q) \mid p \in \mathcal{M}_a, q \in \mathcal{M}_b\}. \tag{2}$$

The properties of product manifolds are provided in Appendix B.1. Based on this framework, the sets $X^{(1)}$ and $X^{(2)}$ from two

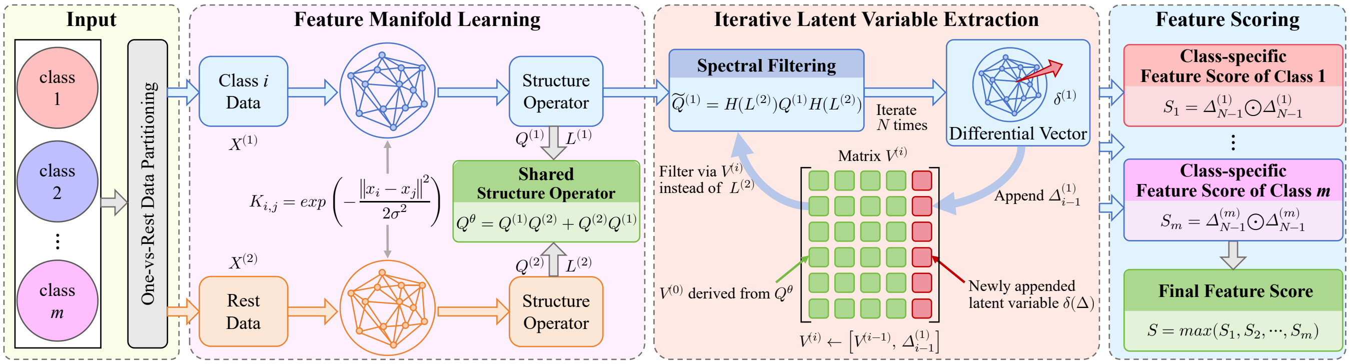

Figure 2: An overview of PRISM. First, we construct a feature association kernel for each class to capture inter-feature dependencies and develop structure operators that characterize the intrinsic geometry of each class manifold. Then, we compute a shared structure operator and implement a spectral filtering mechanism by designing a graph filter to iteratively extract class-specific latent variables. Finally, we use the class-specific latent variable obtained in the final iteration to define the class-specific scores and to construct the final feature score.

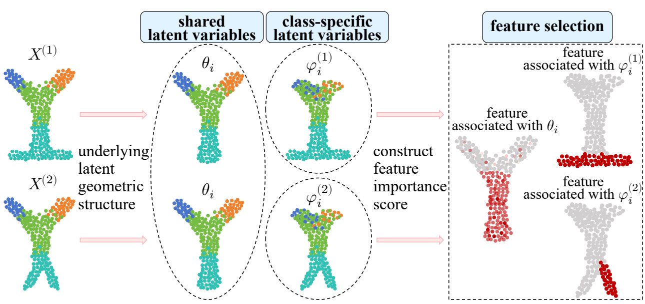

Figure 3: Illustration of our goal: identify features associated with class-specific latent variables.

distinct classes contain observations sampled from two product manifolds $\mathcal{M}_1$ and $\mathcal{M}_2$, respectively, defined as:

$$\mathcal{M}_1 = \mathcal{M}_a \times \mathcal{M}_s, \quad \mathcal{M}_2 = \mathcal{M}_b \times \mathcal{M}_s. \tag{3}$$

Here, $\mathcal{M}_s$ corresponds to the shared latent space between two sets (governed by θ), while $\mathcal{M}_a$ and $\mathcal{M}_b$ correspond to the class-specific latent spaces of $X^{(1)}$ and $X^{(2)}$ (governed by $\varphi^{(i)}$), respectively. We assume that the latent variables $\varphi_i^{(1)}, \varphi_i^{(2)}, \theta_i$ are drawn independently and the observations $x_i^{(1)} \in X^{(1)}, x_i^{(2)} \in X^{(2)}$ follow Eq. 1.

The key challenge in feature selection is that the shared structure (associated with $\mathcal{M}_s$) often dominates the underlying geometry. Under the product manifold framework in Eq. 3, $\mathcal{M}_s$ induces shared geometric (spectral) components across classes, so attenuating these components isolates the class-specific structures (associated with $\mathcal{M}_a$ or $\mathcal{M}_b$). This motivates our design of a spectral filtering mechanism (Sec. 3.4). To implement this mechanism, we first construct feature graphs (Sec. 3.3) to discretely approximate the manifold geometry and enable spectral operations. The background on graph Laplacians is provided in Appendix A.

3.3 Graph Construction and Representation

To implement the proposed product manifold framework in the discrete domain, we first construct the feature manifolds and define a graph operator that explicitly characterizes the shared structure.

3.3.1 Feature Manifold Learning. To capture class-specific differences in the feature associations, we employ a Gaussian kernel defined in Eq. 13[1] to learn the underlying geometric structure of the feature space for each class. Kernel-based approaches are widely employed for nonlinear dimensionality reduction and manifold learning [4, 34]. In contrast to conventional methods that learn the manifold underlying the samples, our method focuses on the manifold structure of features, thereby capturing multivariate associations. For each class $c \in \{1, 2\}$, we compute the affinity matrix $K^{(c)}$ by Eq. 13 and the normalized operator $Q^{(c)}$ by Eq. 14. Then the symmetric normalized Laplacian matrix $L^{(c)}$ is computed by Eq. 15. These operators serve as the discrete approximations of the underlying manifold geometry of the feature space.

3.3.2 The Shared Structure Operator. To separate class-specific structures, we need to identify what is "shared". Let V_s denote the matrix whose columns consist of the eigenvectors in both $Q^{(1)}$ and $Q^{(2)}$ that are associated with shared latent variables θ, and let V_1 and V_2 denote the matrices whose columns consist of the eigenvectors in $Q^{(1)}$ and $Q^{(2)}$ that are associated with class-specific latent variables $\varphi^{(1)}$ and $\varphi^{(2)}$, respectively. In our ideal setting, the two operators $Q^{(1)}$ and $Q^{(2)}$ can be approximated by,

$$Q^{(1)} \approx V_s V_s^T + V_1 V_1^T, \quad Q^{(2)} \approx V_s V_s^T + V_2 V_2^T. \tag{4}$$

To capture shared latent structures, we define the **Shared Structure Operator** Q^{θ} as the symmetric product of $Q^{(1)}$ and $Q^{(2)}$,

$$Q^{\theta} = Q^{(1)}Q^{(2)} + Q^{(2)}Q^{(1)}. \tag{5}$$

We demonstrate the effectiveness of Q^{θ} under the product manifold setting. According to the Lemma 4 in Appendix E.1, the matrices V_s, V_1, V_2 are mutually orthogonal in pairs. It follows that the operator Q^{θ} is equal to,

$$Q^{\theta} \approx 2V_s V_s^T = \sum\nolimits_k v_{0,k}^{(1)} (v_{0,k}^{(2)})^T. \tag{6}$$

Therefore, the leading eigenvectors of Q^{θ} are associated with the shared latent structure in the framework of product manifolds. This operator serves as the target signal we wish to attenuate in the subsequent spectral filtering step.

[1] Eqs. 13–15 are provided in Appendix A.

Algorithm 1 Extraction of a Single Class-Specific Latent Variable

Input: Two sets $X^{(1)} \in \mathbb{R}^{n_1 \times d}$ and $X^{(2)} \in \mathbb{R}^{n_2 \times d}$ from two distinct classes, containing d features with n_1 and n_2 samples respectively. Filter function $h(\lambda) : [0, 1] \to [0, 1]$.
Output: Differential vectors $\delta^{(1)}, \delta^{(2)} \in \mathbb{R}^d$.
Compute the affinity matrices $K^{(1)}, K^{(2)}$ via Eq. 13.
Compute the operators $Q^{(1)}, Q^{(2)}$ and the symmetric normalized Laplacian matrices $L^{(1)}, L^{(2)}$ via Eq. 14 and 15.
Compute the graph filters $H(L^{(1)}), H(L^{(2)})$ using Eq. 7.
Compute the filtered operators $\tilde{Q}^{(1)}, \tilde{Q}^{(2)}$ using Eq. 8.
Compute the differential vectors $\delta^{(1)}, \delta^{(2)}$ from the filtered operators $\tilde{Q}^{(1)}, \tilde{Q}^{(2)}$ respectively.

3.4 Spectral Filtering for Single Class-Specific Latent Variable Extraction

With the graph representations and the shared structure operator established, we now design graph filters to suppress the shared components and extract class-specific latent variables.

To attenuate the shared latent structures that are predominantly associated with the low-frequency components of the Laplacian spectrum, we design a high-pass filter function $H(L^{(c)})$ based on the eigenvalues and eigenvectors of the symmetric normalized Laplacian matrix $L^{(c)}$. Let $\lambda_i^{(1)}, v_i^{(1)}$ and $\lambda_i^{(2)}, v_i^{(2)}$ denote the leading eigenvalues and eigenvectors of $L^{(1)}, L^{(2)}$, respectively. The **Graph Filters** for each class are defined as follows,

$$H(L^{(1)}) = \sum\nolimits_i h(\lambda_i^{(1)}) v_i^{(1)} (v_i^{(1)})^T, \quad H(L^{(2)}) = \sum\nolimits_i h(\lambda_i^{(2)}) v_i^{(2)} (v_i^{(2)})^T, \tag{7}$$

where $h(\lambda) : [0, 1] \to [0, 1]$ is a monotonically increasing function of λ. To extract the class-specific structure of one class (e.g., $X^{(2)}$), we apply the filter of the other class ($X^{(1)}$) to its operator. Thus, we define the **Filtered Operators** $\tilde{Q}^{(1)}$ and $\tilde{Q}^{(2)}$ as:

$$\tilde{Q}^{(2)} = H(L^{(1)}) Q^{(2)} H(L^{(1)}), \quad \tilde{Q}^{(1)} = H(L^{(2)}) Q^{(1)} H(L^{(2)}). \tag{8}$$

The rationale is that $H(L^{(1)})$ suppresses the low-frequency components shared with $L^{(1)}$, thereby revealing the unique structure in $Q^{(2)}$. We refer to the leading eigenvectors of the filtered operators $\tilde{Q}^{(1)}$ and $\tilde{Q}^{(2)}$ as *differential vectors*, denoted by $\delta^{(1)}$ and $\delta^{(2)}$. In Sec. 3.6, we demonstrate that in contrast to the eigenvectors of the unfiltered operators $Q^{(1)}$ and $Q^{(2)}$, under suitable assumptions, these differential vectors are only associated with class-specific latent variables $\varphi^{(1)}$ and $\varphi^{(2)}$. This serves as a theoretical guarantee for the validity and efficacy of our spectral filtering mechanism in disentangling class-specific structures. Algorithm 1 summarizes the procedure for extracting a single class-specific latent variable.

3.5 Iterative Extraction of Class-Specific Latent Variables and Feature Scoring

Typically, data from two different classes within the same dataset may involve multiple class-specific latent variables. In Algorithm 1, we can guarantee that the leading differential vectors of each class, $\delta_0^{(1)}$ and $\delta_0^{(2)}$, are associated exclusively with class-specific latent variables. To guarantee that the subsequent differential vector is non-redundant and associated only with class-specific latent variables that have not yet been identified, we propose an iterative method. Specifically, this method updates the graph filters for the operators $Q^{(1)}, Q^{(2)}$ based on the differential vectors that have been identified. At each iteration, the new graphs for $X^{(1)}$ and $X^{(2)}$ are constructed whose eigenvectors are associated with the shared latent variables, the identified class-specific latent variables, and all their cross-products. Thus, we can design the new graph filters to attenuate all these eigenvectors, thereby enabling the discovery of new class-specific latent variables. Algorithm 2 summarizes the procedure for extracting the multiple class-specific latent variables.

Algorithm 2 Iterative Extraction of Multiple Class-Specific Latent Variables and Feature Scoring

Input: Two sets $X^{(1)} \in \mathbb{R}^{n_1 \times d}$ and $X^{(2)} \in \mathbb{R}^{n_2 \times d}$ from two distinct classes, containing d features with n_1 and n_2 samples respectively. Number of iterations N. Filter function $h(\lambda) : [0, 1] \to [0, 1]$. The dimension of shared latent space k_0.
Output: Differential vectors $\Delta_0^{(1)}, \ldots, \Delta_{N-1}^{(1)} \in \mathbb{R}^d$. Class-specific feature scores S_1, S_2. Overall feature score S.
Compute $\delta_0^{(1)}$ as $\Delta_0^{(1)}$ via Algorithm 1.
Compute Q^{θ} via Eq. 5 and its leading eigenvectors $V^{(0)} \in \mathbb{R}^{d \times k_0}$.
for $i = 1$ **to** $N - 1$ **do**
  Concatenate $V^{(i)} \leftarrow \left[V^{(i-1)}, \Delta_{i-1}^{(1)} \right]$.
  Compute $\Delta_i^{(1)}$ via Algorithm 1 with inputs $X^{(1)}$ and $V^{(i)}$.
end for
Compute the class-specific feature scores S_1, S_2 using Eq. 9.
Compute the overall feature score S using Eq. 10.

Proposed Feature Score. Each differential vector extracted by Algorithm 2 is d-dimensional, consistent with the feature dimension of sets $X^{(1)}, X^{(2)}$. Consider the differential vectors $\Delta_{N-1}^{(1)}, \Delta_{N-1}^{(2)}$ of $X^{(1)}, X^{(2)}$ obtained in the N-th iteration, we define the **class-specific feature scores** for the two sets as S_1 and S_2, respectively,

$$S_1 = \Delta_{N-1}^{(1)} \odot \Delta_{N-1}^{(1)}, \quad S_2 = \Delta_{N-1}^{(2)} \odot \Delta_{N-1}^{(2)}, \tag{9}$$

where $\odot$ denotes the Hadamard product. These two scores respectively indicate the importance of each feature in capturing the distinctive structures of $X^{(1)}$ and $X^{(2)}$. Finally, we integrate them to obtain the **overall feature score**, denoted by

$$S = \max(S_1, S_2), \tag{10}$$

where the maximum is taken element-wise. The proposed feature score S effectively reflects the contribution of each feature in distinguishing the differential structure within the feature space of $X^{(1)}$ and $X^{(2)}$, thereby enabling the identification of features with high discriminative capability. Our proposed method PRISM can be extended to multi-class classification problems. The details are provided in Appendix B.2.

3.6 Theoretical Guarantee

In this part, we provide the theoretical guarantee for the proposed method. Our goal is to demonstrate that the differential vector extracted by our spectral filtering mechanism indeed converges to the leading eigenfunction of the class-specific manifold.

We begin with the convergence results of the eigenvectors of the discrete Laplacian to the eigenfunctions of the Laplace-Beltrami (LB) operator under the manifold setting. Generalizing the Euclidean Laplacian to the Riemannian manifold, the LB operator acts as the continuous graph Laplacian governing the geometric structure of the underlying manifold. Theorem 2 in Appendix E.1 implies that the eigenvectors of the random walk Laplacian converge to the eigenfunctions of the LB operator at a certain rate. Based on this theorem, we derive a similar bound for the eigenvectors of the symmetric normalized Laplacian. See the Theorem 3 in Appendix E.1. Theorems 2 and 3 demonstrate the convergence of the eigenvectors of the graph Laplacian under a single manifold corresponding to a single set. However, we consider the case of two sets in our work. Thus, we extend the spectral analysis to the product manifold setting involving two sets and bridge the theoretical gap. The detailed theoretical foundation of our proof is provided in Appendix E[2]. In the following, we show that the similar convergence result holds under the product manifold setting associated with two sets.

Theorem 1. *Let $X^{(1)}, X^{(2)}$ be two sets of d observations sampled uniformly at random from the product manifolds $\mathcal{M}_1 = \mathcal{M}_a \times \mathcal{M}_s, \mathcal{M}_2 = \mathcal{M}_b \times \mathcal{M}_s$ respectively. Let*

$$\delta^{(2)} = \underset{\|v\|=1}{\arg\max}\; v^T P^{(1)} Q_\tau^{(2)} P^{(1)} v \tag{11}$$

be the differential vector obtained in step 7 of Algorithm 1. Under the assumptions (i)-(iii) for both manifolds, as $d \to \infty$, with probability at least $1 - 4m^2 d^{-10} - (2m+6)d^{-9}$, the following holds

$$\left\|\delta^{(2)} - \frac{\alpha}{\sqrt{pd}}\beta_{\pi_b(X)}(f_1^{(b)})\right\|^2 \leq O(m\sigma_d) + O\left(m\sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right). \tag{12}$$

Theorem 1 provides a convergence guarantee for Algorithm 1. It demonstrates that given a sufficiently large number of features, the leading differential vector $\delta^{(2)}$ of the filtered operator $\tilde{Q}^{(2)}$ obtained in step 7 of Algorithm 1 converges to the leading eigenfunction of $\mathcal{M}_b$ (the class-specific manifold). Thus, the differential vector captures the leading class-specific latent variable that is not shared between the two sets, proving that our filtered operator successfully isolates the class-specific structure from the shared structure.

4 Experiments

We evaluate the performance of PRISM in comparison with both commonly used and state-of-the-art feature selection methods on a range of synthetic and real-world datasets. In all experiments, the data are split into training and testing sets, and a nested cross-validation strategy is adopted to ensure fair and reliable evaluation. All competing FS methods are carefully tuned to achieve their best performance on the training set. Additional details and results are provided in Appendix C and D, respectively. The source code is available at https://github.com/Lakers-memory/PRISM.

4.1 XOR-100 Problem

To evaluate the capability of PRISM in capturing non-linear feature interactions, we generate a synthetic XOR dataset consisting of $d = 100$ binary features and $N = 50$ samples [43]. Each feature is independently drawn from a Bernoulli distribution, and class labels are assigned based on the XOR operation between two predefined features (f_1 and f_5), making only these two features informative, while the remaining 98 features serve as irrelevant noise. This setup poses a significant challenge to conventional filter methods that rely solely on univariate statistical significance.

We conduct 200 Monte Carlo simulations of data generation. Figure 4 presents the statistical distributions of normalized feature scores for each method. In each simulation, the top two features with the highest scores are selected, and the average of correct selections is shown in parentheses. The results demonstrate that PRISM consistently identifies the interaction between f_1 and f_5 across all simulations, whereas all other compared methods fail to do so, except for ManiFeSt and ReliefF.

Notably, PRISM and ManiFeSt both achieve a perfect average number of correct selections of 2 in identifying the two informative features, with ManiFeSt likewise employing a manifold learning approach in the feature space. Although ReliefF is a univariate method, it indirectly incorporates multivariate information via the local geometry of nearest-neighbor samples and thus also recovers these two features. However, its average of correct selections is only 0.765. This performance gap further underscores the superior ability of PRISM to capture multi-feature interactions.

4.2 Madelon

We evaluate the robustness of PRISM to small sample size on the Madelon dataset [14]. This dataset contains 2600 samples with 500 features, including 5 hypercube coordinates, 15 of their linear combinations, and 480 independent Gaussian noise features. The data is partitioned into train and test sets using 10-fold cross-validation. To assess performance in small-sample scenarios, we consider two settings: (i) the full train set of 2340 samples, (ii) only 5% of the train set, comprising 117 samples. In both settings, we evaluate performance by training an SVM classifier on the full train set.

Figure 5 shows that when feature selection is performed using the full train set, all methods can identify relevant features. PRISM achieves the highest average test accuracy of 91.92%, outperforming ManiFeSt (90.84%) and ReliefF (90.73%). When feature selection is conducted using only 5% of the train set, only PRISM and ManiFeSt can identify the relevant features, while all other competing methods fail. In this setting, PRISM again achieves the best performance with a test accuracy of 91.50%, exceeding ManiFeSt's 90.58%. These results show that PRISM exhibits remarkable robustness to limited sample sizes and strong generalization capability. Additional results on varying sample sizes, which further validate this robustness of PRISM, are detailed in Appendix D.2.

4.3 Lung

We evaluate the result reusability of PRISM on the Lung dataset, which contains 203 samples, 3312 features, and 5 classes. We first isolate the data from Classes 1 and 2 and perform feature selection separately using PRISM and ManiFeSt. A 10-fold cross-validation procedure is then repeated five times, and the features selected in each iteration are recorded. The average classification accuracies obtained with the top 10, 50, 100, 150, 200, 300, and 500 selected features are reported in Figure 6a. Compared with ManiFeSt, PRISM not only produces overall feature scores for Classes 1 and 2 but also, based on the extracted class-specific latent variables, yields class-specific feature scores for each class (see Eq. 9). This allows PRISM to identify features that are unique to each class. Leveraging this property, when new classes emerge, previously selected global

[2]Due to space limitations, we only provide the main sketch of proofs, while the detailed proofs can be found in Appendix E of our GitHub repository at https://github.com/Lakers-memory/PRISM.

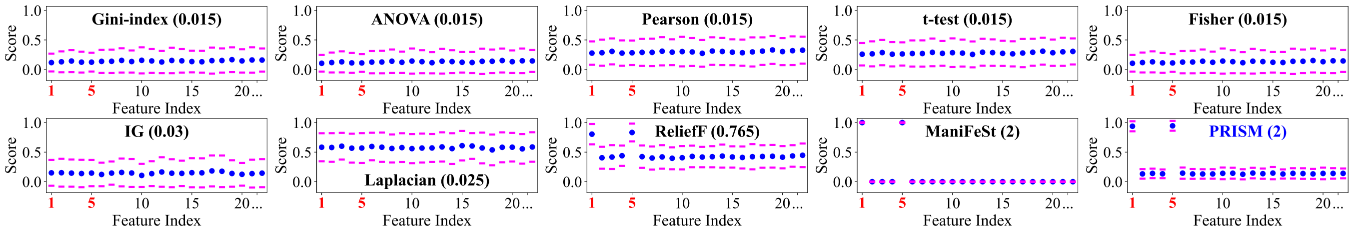

Figure 4: Feature score for the XOR-100 problem. Blue dots denote the average score, and purple lines indicate the standard deviation.

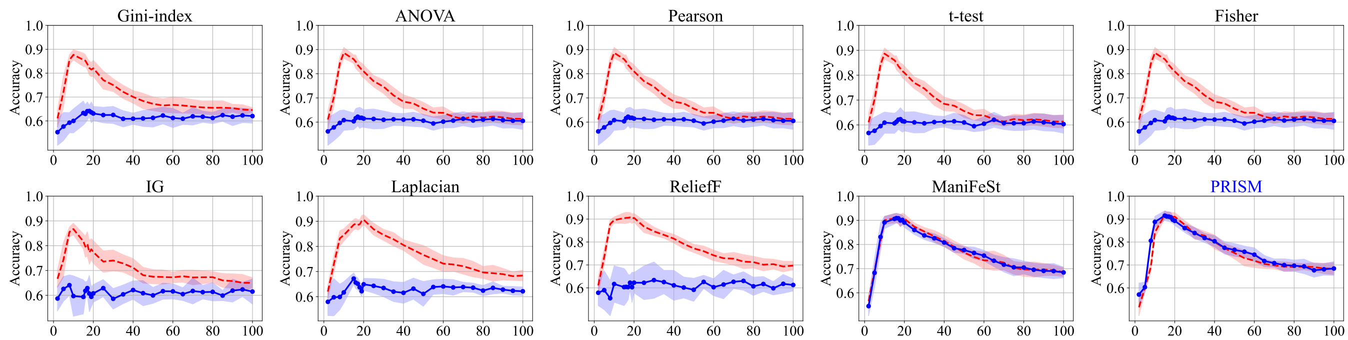

Figure 5: Test accuracy vs. number of features on the Madelon dataset. Lines denote the average test accuracy, and the shaded area denote the standard deviation. The red dashed line denotes feature selection using the full training set, while the blue solid line uses only 5% of the data.

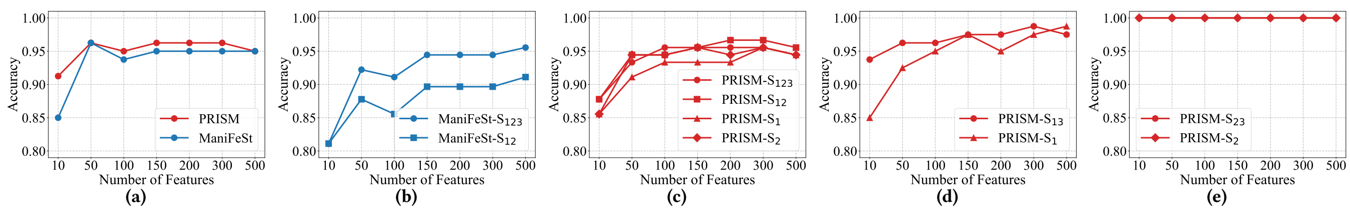

Figure 6: Accuracy under different settings on the Lung dataset. Suffixes S_{123}, S_{12}, S_1, S_2, and S_{13}/S_{23} correspond to settings (i)–(v), respectively.

and class-specific features can be reused. To verify the reusability of the feature selection results obtained by PRISM, we consider the case in which data from Class 3 of the Lung dataset are added.

For PRISM, we contrast a full retraining benchmark (Setting i), where feature selection is performed anew on all three classes, against three feature reuse strategies. These strategies include utilizing previously selected global features (Setting ii) or reusing previously selected class-specific features of Class 1 (Setting iii) and Class 2 (Setting iv). The results are shown in Figure 6c. We observe that, after reusing the feature selection results from Classes 1 and 2 in Settings (ii)-(iv), the classification accuracies remain very close to those obtained in Setting (i), and in fact Setting (ii) yields slightly higher accuracies than Setting (i) when 200, 300, or 500 features are used. For ManiFeSt, we conduct the same comparison between Settings (i) and (ii), and the results are shown in Figure 6b. Notably, the performance in Setting (ii) drops substantially compared to Setting (i), further indicating that PRISM exhibits markedly superior result reusability compared to ManiFeSt.

To further demonstrate the discriminative power of the class-specific features identified by PRISM when new classes emerge, we compare feature selection anew (Setting v) against reusing previously selected class-specific features (Setting iii or iv) to distinguish Class 1 (or 2) from Class 3. The results in Figure 6d and Figure 6e show that the accuracies in these two settings are very close, and in some cases almost overlap, indicating that the class-specific features selected by PRISM possess strong discriminative capability.

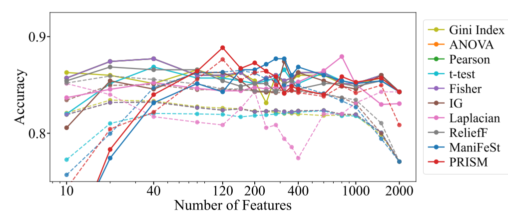

Figure 7: Accuracy on the colon cancer dataset. The dashed and solid lines represent the average test and validation accuracy, respectively.

4.4 Colon Cancer

We evaluate the generalization capability of PRISM on a colon cancer gene expression dataset [2]. The dataset comprises 62 tissue samples, each with expression levels for 2000 genes. Among them, 22 samples are from normal tissues and 40 from colon cancer tissues. The data are split into 90% for training and 10% for testing, and the results are averaged over 50 cross-validation iterations.

Figure 7 presents the average classification accuracy across different feature subsets for various feature selection methods. PRISM achieves the highest test accuracy of 87.65%, outperforming all competing methods, the best of which attains a lower accuracy of 86.51%. Notably, PRISM demonstrates superior generalization capability compared to all competing methods, as evidenced by the smaller gap between its validation and test accuracies.

Table 1: Accuracy (%) and standard deviation for different variants in the ablation study on benchmark datasets.

Method	ALLAML (7129/72)	Arcene (10000/200)
wo HPF-1	82.50 ± 15.51	67.80 ± 6.79
wo HPF-2	79.00 ± 6.27	65.40 ± 7.34
wo ILVE	71.50 ± 11.81	61.25 ± 6.56
wo SSO	78.00 ± 3.71	64.50 ± 4.12
wo INTG-1	94.50 ± 6.94	76.40 ± 8.28
wo INTG-2	94.50 ± 6.94	75.80 ± 4.96
PRISM	**98.50 ± 2.24**	**80.00 ± 7.91**

4.5 Ablation Study

We conduct this experiment to study the impact of the high-pass filter function, iterative latent variable extraction, shared structure operator and feature scoring mechanism on the performance of PRISM. We implement four variants of PRISM. **wo HPF** removes the high-pass filter function in PRISM. First, we substitute the high-pass filter function with a low-pass filter, denoted as **wo HPF-1**. Second, we remove the spectral filtering step, denoted as **wo HPF-2**. **wo ILVE** removes the iterative latent variable extraction step and directly employs the first extracted latent variable for the subsequent feature scoring. **wo SSO** uses structure operators to iteratively extract latent variables instead of using the shared structure operator in Eq. 5. For example, we replace Q^{θ} with $Q^{(2)}$ in Step 4 of Algorithm 2. **wo INTG** directly uses the individual class-specific scores as the final feature score instead of performing the integration described in Eq. 10. Specifically for binary classification tasks, we evaluate two variants: **wo INTG-1**, which utilizes only S_1, and **wo INTG-2**, which relies solely on S_2.

We randomly choose two datasets to conduct this experiment, which are ALLAML and Arcene. Detailed dataset descriptions are in Appendix C. We average the performance obtained with the first 10, 50, 100, 150, and 200 selected features. Table 1 shows the overall experiment results. We observe that the performance of PRISM significantly outperforms all ablated variants. Specifically, the significant performance drop observed in **wo HPF** demonstrates that the high-pass filter is essential for attenuating shared low-frequency signals and amplifying discriminative class-specific variations (high-frequency), a capability that low-pass filters fail to provide. Furthermore, the degradation in **wo ILVE** highlights the complexity of high-dimensional data, where discriminative information is distributed across multiple latent dimensions that require iterative extraction to be fully captured. The poor results of **wo SSO** validate the necessity of the shared structure operator. Without this operator, PRISM loses the ability to distinguish class-specific features from the shared structure. Finally, the comparison with **wo INTG** confirms that while individual class-specific scores are informative, integrating them via the proposed max-pooling mechanism maximizes discriminative power by preserving the unique characteristics of every class.

4.6 Hyperparameter analysis

In this part, we investigate how hyperparameters influence the effectiveness of PRISM, focusing on four key parameters: the index of the K-th nearest neighbor used to estimate the local kernel bandwidth in affinity matrix construction, the number of leading eigenvectors k used in the graph filters, the number of leading eigenvectors k_0 used in the shared structure operator, and the number of iterations N. For efficient tuning across datasets of varying dimensions, we normalize K, k, and k_0 with respect to the feature dimensionality, resulting in ratios that lie within the range $[0, 1]$. Figure 8 illustrates the classification accuracy on the ALLAML dataset under varying parameter settings. Specifically, for the local neighborhood size K, accuracy peaks within the range of $[0.05, 0.1]$; values below this threshold suffer from noise due to graph sparsity, while excessive values lead to oversmoothing that blurs the distinct local manifold structures. Regarding the spectral dimensions k and k_0, we observe a consistent performance improvement as their ratios approach $[0.9, 1.0]$, confirming that retaining nearly full spectral information is critical for capturing subtle high-frequency class-specific variations and constructing a comprehensive shared structure. Finally, the performance stabilizes around $N = 5$, indicating that five iterations are sufficient to fully extract the discriminative latent variables without incurring unnecessary computational overhead.

4.7 Additional Results

We further compare PRISM with different FS methods, including filter-based approaches and recent deep learning-based methods such as E2E-FS [7], LassoNet [23], GradEnFS [26], Skinny Trees [18], and SAND [28], on seven additional datasets, all of which exhibit pronounced high-dimensional small-sample characteristics. Among these, the Lymphoma dataset contains 9 classes, and CLL_SUB_111 contains 3 classes, while the remaining datasets are binary classification problems. Detailed dataset descriptions are in Appendix C. We average the performance obtained with the first 10, 50, 100, 150, and 200 selected features. The results are shown in Table 2. We observe that PRISM obtains the highest classification accuracies for all datasets, further demonstrating its superior performance in high-dimensional small-sample settings. In addition, the complexity analysis of PRISM is provided in Appendix D. Runtime and memory comparisons with the strongest baselines are shown in Table 3.

5 Conclusion

In this work, we propose a novel supervised feature selection method which is termed PRISM. PRISM integrates product manifold with spectral graph analysis for feature-space manifold learning, explicitly modeling multivariate feature interactions and implementing a spectral filtering mechanism to capture class-specific latent variables representing the underlying differential structure. We provide an asymptotic convergence analysis that theoretically guarantees the feasibility of uncovering class-specific discriminative structures from the shared structure. Comprehensive experiments on diverse benchmark datasets demonstrate that our method consistently outperforms state-of-the-art baselines, significantly enhancing generalization capability and robustness in high-dimensional small-sample settings, while crucially ensuring superior feature reusability when new classes emerge.

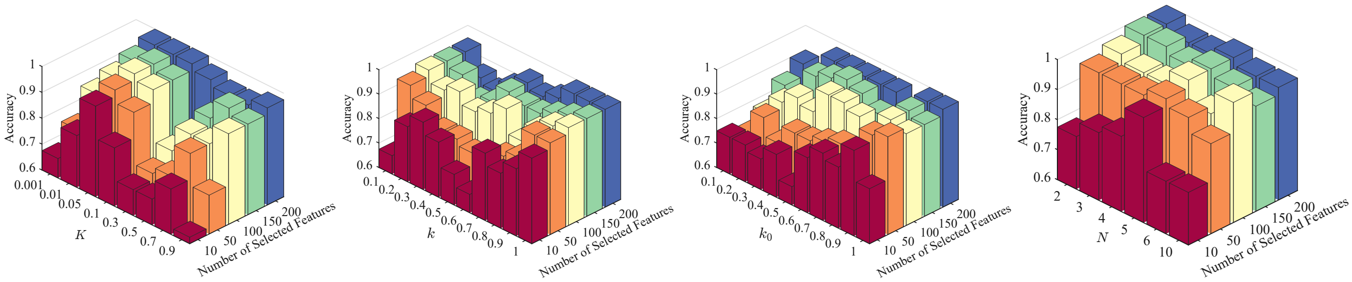

Figure 8: Hyperparameter sensitivity analysis on the ALLAML dataset.

Table 2: Comparison of accuracy (%) and standard deviation for various FS methods on benchmark datasets.

Method	Prostate (5966/102)	ALLAML (7129/72)	Arcene (10000/200)	SMK_CAN_187 (19993/187)	GLI_85 (22283/85)	Lymphoma (4026/96)	CLL_SUB_111 (11340/111)
No FS	91.82 ± 0.00	97.50 ± 0.00	83.00 ± 0.00	77.89 ± 0.00	84.44 ± 0.00	94.00 ± 0.00	68.33 ± 0.00
Gini Index	93.18 ± 0.64	96.00 ± 3.35	74.40 ± 4.83	75.37 ± 2.31	84.89 ± 0.99	79.20 ± 21.34	68.67 ± 2.47
ANOVA	93.63 ± 1.07	97.50 ± 3.06	66.80 ± 3.96	76.42 ± 2.94	86.22 ± 0.99	90.80 ± 7.01	53.67 ± 3.98
Pearson	93.63 ± 1.07	97.50 ± 3.06	66.80 ± 3.96	76.42 ± 2.94	86.22 ± 0.99	91.20 ± 7.56	54.00 ± 4.80
Fisher	93.63 ± 1.07	97.50 ± 3.06	66.80 ± 3.96	76.42 ± 2.94	86.22 ± 0.99	90.80 ± 7.01	53.67 ± 3.98
IG	93.21 ± 0.81	97.50 ± 3.54	73.40 ± 5.55	71.37 ± 3.60	87.56 ± 2.53	94.40 ± 3.58	70.21 ± 3.56
Laplacian	91.52 ± 1.39	91.00 ± 2.24	69.80 ± 4.97	72.84 ± 3.52	71.56 ± 6.55	92.00 ± 8.12	57.33 ± 7.87
ReliefF	92.24 ± 0.73	97.50 ± 3.06	67.40 ± 7.57	77.47 ± 5.35	87.11 ± 2.43	94.40 ± 3.85	59.67 ± 3.98
Inf-FS	93.45 ± 0.92	97.50 ± 3.06	71.25 ± 7.50	78.95 ± 2.27	86.51 ± 2.48	89.20 ± 14.46	68.96 ± 3.55
ILFS	92.78 ± 0.88	94.00 ± 8.22	69.25 ± 6.14	79.63 ± 2.88	88.42 ± 3.26	83.60 ± 12.76	70.58 ± 2.54
E2E-FS	93.85 ± 0.82	88.50 ± 7.20	71.80 ± 6.69	76.11 ± 3.90	76.44 ± 3.37	83.20 ± 13.97	61.67 ± 4.41
LassoNet	91.15 ± 4.91	93.00 ± 1.12	65.00 ± 10.56	74.05 ± 3.62	83.62 ± 3.32	81.20 ± 17.87	56.32 ± 3.66
GradEnFS	90.79 ± 4.58	85.00 ± 8.84	63.40 ± 4.16	71.11 ± 5.54	78.35 ± 3.63	91.60 ± 6.84	62.50 ± 4.63
Skinny Trees	89.15 ± 6.13	83.50 ± 8.40	63.20 ± 9.07	70.68 ± 3.54	76.78 ± 4.15	86.00 ± 12.41	58.72 ± 4.22
SAND	95.33 ± 0.70	92.50 ± 0.00	64.80 ± 9.81	75.53 ± 2.47	84.67 ± 4.28	92.40 ± 4.77	56.23 ± 3.17
ManiFeSt	94.10 ± 0.96	95.00 ± 3.06	75.40 ± 4.77	80.23 ± 3.86	88.89 ± 3.11	88.00 ± 9.52	72.36 ± 2.16
PRISM (ours)	**95.72 ± 0.56**	**98.50 ± 2.24**	**80.00 ± 7.91**	**83.37 ± 3.34**	**90.15 ± 2.67**	**94.60 ± 4.78**	**75.89 ± 2.64**

Table 3: Runtime (s) and peak GPU memory in parentheses (MiB) of different FS methods on benchmark datasets.

Method	Prostate	ALLAML	GLI_85	Lymphoma
SAND	87.7 (22158)	85.7 (22166)	89.0 (22150)	89.0 (22164)
ManiFeSt	4.3 (3808)	7.2 (5442)	181.6 (20210)	224.7 (4352)
PRISM (ours)	36.1 (6412)	60.8 (8936)	473.1 (22132)	138.1 (2904)

Acknowledgments

This work is supported by the Brain Science and Brain-like Intelligence Technology-National Science and Technology Major Project (Grant No. 2021ZD0201302), the National Natural Science Foundation of China (Grant No. 62388101), the Fundamental Research Funds for the Central Universities, and Beijing Advanced Innovation Center for Future Blockchain and Privacy Computing. This work is also supported by the A*STAR under its IAF-ICP (H25-MCP3438) and Ministry of Education, Singapore, under its Academic Research Fund (AcRF) Tier 1 grant, and funded through the SMU-SUTD Internal Research Grant Call (SMU-SUTD 2023_02_01).

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A Graph Laplacian and Graph Representation

Consider a data matrix $X^{(c)} \in \mathbb{R}^{n_c \times d}$ containing n_c samples belonging to a specific class c. The rows of the matrix correspond to observations that satisfy the double manifold assumption of Eq. 1. Let $x_i^{(c)}$ denote the i-th observation in $X^{(c)}$. An affinity matrix $K^{(c)}$ is computed using the Gaussian kernel with the Euclidean norm,

$$K_{i,j}^{(c)} = \exp\left(-\|x_i^{(c)} - x_j^{(c)}\|^2 / 2\sigma_c^2\right), \tag{13}$$

where σ_c is the bandwidth parameter that controls the decay of the Gaussian kernel.

Let $D^{(c)}$ be a diagonal matrix whose elements are row sums of $K^{(c)}$. We compute the operator by,

$$Q^{(c)} = (D^{(c)})^{-1/2} K^{(c)} (D^{(c)})^{-1/2}. \tag{14}$$

And we denote the symmetric normalized Laplacian matrix by,

$$L^{(c)} = I - Q^{(c)}. \tag{15}$$

An important property of the Laplacian matrix is that the eigenvectors corresponding to its large eigenvalues can effectively capture the underlying geometric structure of the data. Thus, we analyze the graph Laplacian $L^{(c)}$ to capture the specific structure of $X^{(c)}$.

B Methodological Details

B.1 Properties of product manifolds

Property 1 (Function Extension). *Let $\pi_a : \mathcal{M} \to \mathcal{M}_a$ and $\pi_b : \mathcal{M} \to \mathcal{M}_b$ be the canonical projection operators that map a point in $\mathcal{M}$ to its corresponding points in $\mathcal{M}_a$ and $\mathcal{M}_b$, respectively. Then, the projection operator can be used to extend a real function $f_a : \mathcal{M}_a \to \mathbb{R}$ on $\mathcal{M}_a$ to a function $f : \mathcal{M} \to \mathbb{R}$ over the product $\mathcal{M}$ by $f(x) = f_a(\pi_a(x))$.*

Property 2 (Spectral Decomposition). *Let $\mu_i^{(j)}, f_i^{(j)}$ denote the i-th smallest eigenvalue and corresponding eigenfunction of the Laplace-Beltrami operator for manifold $\mathcal{M}_j$. Let $\mu_{l,k}^{(1)}$ denote the (l,k)-th smallest eigenvalue of $\mathcal{M}_1$. If $\mu_{l,k}^{(1)}$ is the i-th smallest eigenvalue of $\mathcal{M}_1$,* *this eigenvalue has the subscript (l, k). According to [47], the spectral components of the product manifolds $\mathcal{M}_1$ and $\mathcal{M}_2$ satisfy the following decomposition properties:*

(i) ***Eigenfunctions (Pointwise Product).*** *The eigenfunctions are equal to the pointwise product of component eigenfunctions,*

$$
\begin{aligned}
f_{l,k}^{(1)}(x) &= f_l^{(a)}(\pi_a(x)) \cdot f_k^{(s)}(\pi_s(x)), \\
f_{m,k'}^{(2)}(x) &= f_m^{(b)}(\pi_b(x)) \cdot f_{k'}^{(s)}(\pi_s(x)).
\end{aligned} \tag{16}
$$

(ii) ***Eigenvalues (Linear Sum).*** *The corresponding eigenvalues are the sums of the component eigenvalues,*

$$
\mu_{l,k}^{(1)} = \mu_l^{(a)} + \mu_k^{(s)}, \quad \mu_{m,k'}^{(2)} = \mu_m^{(b)} + \mu_{k'}^{(s)}. \tag{17}
$$

B.2 Multi-class PRISM extension

In the method of our paper, we concentrate on binary classification problems, as they serve as the standard stepping-stone to multi-class feature selection [19]. In fact, our method can be naturally extended to multi-class problems. Here, we introduce two strategies for extending our method to the multi-class setting. Both approaches follow the same core steps as outlined in Algorithm 2.

The first strategy is a theory-grounded extension that aims to identify a shared structure operator among multiple classes. As shown in Eq. 5 in Section 3.3 of the paper, we define a shared structure operator for two classes. Under the assumption of product manifolds, this formulation can be generalized to the multi-class case. Specifically, for a classification problem with m classes, we can construct a shared structure operator denoted by

$$
Q^{\theta} = \sum_{1 \le i < j \le m} Q^{(i)} Q^{(j)} + Q^{(j)} Q^{(i)}, \tag{18}
$$

which captures the underlying structure common to all classes. Assuming that the data from the m classes satisfy the product manifold setting pairwise, then according to Eq. 6, $Q^{(i)}Q^{(j)} + Q^{(j)}Q^{(i)}$ captures the shared structure between class i and class j. Consequently, Q^{θ} in Eq. 18 captures the shared structure across all pairwise combinations of class data. Based on this shared structure, we compute the class-specific score for each class by pairing the data from the i-th class with the shared structure operator Q^{θ} as input to Algorithm 2. This process is repeated for all m classes, yielding a set of class-specific scores $\{S_1, S_2, \ldots, S_m\}$. Finally, we aggregate these scores to obtain the overall feature importance score, denoted by

$$
S = \max(S_1, S_2, \ldots, S_m), \tag{19}
$$

where the maximum is taken element-wise. This strategy allows our method to capture both shared and class-specific structures in the data, making it well-suited for multi-class problems. However, a limitation of this strategy is that the computational cost increases significantly with the number of classes. To address this issue, we propose a second strategy as a practical approximation.

The second strategy addresses this issue by adopting a one-vs-rest approach. Specifically, for each class i, we divide the dataset into two parts: the samples belonging to class i, and those belonging to all other classes. These two subsets are then used as input to Algorithm 2, following the same iterative procedure to compute a class-specific score S_i that captures the discriminative structure of class i against the rest. By repeating this process for all m classes, we obtain class-specific scores $\{S_1, S_2, \ldots, S_m\}$, which are then aggregated according to Eq. 19. This strategy is more computationally efficient and scales better with the number of classes, while still preserving class-specific information. We adopt this strategy for extending our method to the multi-class setting. Algorithm 3 summarizes the procedure for the multi-class extension of PRISM.

Algorithm 3 Multi-class PRISM Extension

1: **Input:** m datasets $\{X^{(i)} \in \mathbb{R}^{n_i \times d}\}_{i=1}^{m}$, each containing d features with n_i samples in the i-th dataset. Number of iterations N. Filter function $h(\lambda) : [0, 1] \to [0, 1]$. The dimension of shared latent space k_0.
2: **Output:** Class-specific feature scores $\{S_1, S_2, \ldots, S_m\}$. Feature score S.
3: **for** $i = 1$ **to** m **do**
4:     Construct two subsets: $X^{(i)}$ and $X_{\text{rest}}^{(i)} = \bigcup_{j=1, j \ne i}^{m} X^{(j)}$.
5:     Compute the class-specific feature score S_i via Algorithm 2 with inputs $X^{(i)}$ and $X_{\text{rest}}^{(i)}$.
6: **end for**
7: Compute the feature score S using Eq. 19.

C Experiments - Implementation Details

In all experiments, we adopt a nested cross-validation strategy, where the training set is further split using 10-fold cross-validation to optimize model performance. To ensure robustness on small datasets, the cross-validation process is repeated with randomly shuffled samples. All procedures, including data normalization, feature selection, and SVM hyperparameter tuning, are strictly confined to the training folds to avoid any risk of data leakage.

To enhance the generalization capability of the model, SVM hyperparameters are optimized based on validation data. In contrast, feature selection (FS) method hyperparameters are tuned using only the training data within the inner cross-validation procedure. This decoupled tuning scheme is adopted to prevent joint optimization of SVM and FS hyperparameters, which can lead to overfitting or confounding effects. FS hyperparameter tuning is carried out with respect to varying feature subset sizes. As shown in Table 2, we search for optimal hyperparameters using the training data separately for each candidate number of features. The selected configuration is then fixed and evaluated once on the held-out test set, and the resulting test performance is reported.

Data normalization. Consistent with the approach used by Atashgahi et al. [3], the features in the Madelon dataset are normalized by centering around the mean and scaling to unit variance. No normalization is applied to the remaining datasets, as their feature distributions do not necessitate such treatment.

FS hyperparameter tuning. For ReliefF and for IG which extend the classic Information Gain score to continuous features via a k nearest neighbors approach, we tune k over the grid $\{1, 3, 5, 10, 15, 20, 30, 50\}$. For Laplacian Score, which requires a kernel scale parameter, we evaluate scales corresponding to the $\{1, 5, 10, 30, 50, 70, 90, 95, 99\}$ percentiles of the Euclidean distance distribution. ManiFeSt involves only a single scale parameter σ_ℓ. For the XOR and Madelon datasets, due to the pronounced feature interaction patterns, σ_ℓ is fixed at 0.1 times the median of Euclidean distances without further tuning. For all other datasets, σ_ℓ is tuned over the $\{5, 10, 30, 50, 70, 90, 95\}$ percentiles, with additional evaluations at the 1st and 99th percentiles to capture nonlocal interactions. The PRISM method involves four key hyperparameters: the index of the K-th nearest neighbor used to estimate the local kernel bandwidth in kernel matrix construction, the number of leading eigenvectors k used in the filtering operator, the number of leading eigenvectors k_0 used in the shared structure operator, and the number of iterations N. The parameter K is tuned within a range corresponding to 5% to 10% of the feature dimension, while k and k_0 are selected from a range spanning 90% to 100% of the feature dimension. The number of iterations N is typically set to 5.

SVM hyperparameter tuning. The hyperparameter optimization follows the classical procedure proposed by Hsu et al. [17], conducting a grid search over exponentially spaced values for both the penalty parameter $C = \{2^{-5}, 2^{-2}, 2^{1}, 2^{4}, 2^{7}, 2^{10}, 2^{13}\}$ and the kernel scale $\gamma = \{2^{-15}, 2^{-12}, 2^{-9}, 2^{-6}, 2^{-3}, 2^{0}, 2^{3}\}$.

Experimental setup. All experiments were conducted in a Python environment equipped with an RTX 4090 GPU (24GB memory). To accelerate computation, GPU-optimized libraries such as *cupy* and *cuML* were utilized throughout the experiments.

Implementation of baseline methods. The implementations of the baseline feature selection methods are as follows: IG and ANOVA were implemented using the *scikit-learn* library. Gini Index, t-test, Fisher Score, Laplacian Score, and ReliefF were implemented using the skfeature package developed by Arizona State University [24]. Pearson correlation was computed using the built-in function from Pandas. Inf-FS [33], ILFS [32], E2E-FS [7], LassoNet [23], GradEnFS [26], Skinny Trees [18], SAND [28] and ManiFeSt [9] were implemented following the algorithms detailed in their original papers or utilizing the source code provided by the authors.

Details on the hypercube dataset. In the experiment described in Section D.3, we construct a high-dimensional dataset based on a hypercube embedded in a 10-dimensional space. Specifically, four Gaussian clusters are generated at the vertices of the hypercube, producing a total of 2000 samples. These clusters are arbitrarily grouped into two classes, each containing two clusters, to form a binary classification task. To increase the dimensionality, the original 10-dimensional data are mapped to a 200-dimensional space by appending random noise to the remaining 190 dimensions. Only the first 10 dimensions carry informative features, while the rest are purely noisy.

Table 4: Datasets used in the experiments.

Dataset	#samples	#features	#classes	type
Colon	62	2000	2	Biology
Prostate	102	5966	2	Biology
ALLAML	72	7129	2	Biology
Arcene	200	10000	2	Mass Spectrometry
SMK_CAN_187	187	19993	2	Biology
GLI_85	85	22283	2	Biology
Lung	203	3312	5	Biology
Lymphoma	96	4026	9	Biology
CLL_SUB_111	111	11340	3	Biology
Madelon	2600	500	2	Artificial
Gisette	7000	5000	2	Digit recognition
RELATHE	1427	4322	2	Text

Description of benchmark datasets. We evaluate our method on a diverse collection of high-dimensional datasets, primarily focusing on biological and medical domains. In addition to these biomedical datasets, we utilize Gisette and RELATHE to assess performance in other high-dimensional scenarios. The Gisette dataset, derived from the NIPS 2003 feature selection challenge, includes 5000-dimensional feature vectors for 7000 samples to distinguish between the handwritten digits "4" and "9". RELATHE is a benchmark dataset in the text domain, which is highly sparse and comprises 4322 features and 1427 samples. All datasets employed in this study are publicly available at https://jundongl.github.io/scikit-feature/datasets.html. Table 4 shows details of the datasets used. All experiments adopt a 9:1 train-test split strategy. For the Prostate cancer dataset, performance is averaged over 30 independent cross-validation iterations, while for Arcene and the remaining biological datasets, we employ 5 iterations to ensure statistical reliability.

D Additional Results

D.1 Complexity analysis of PRISM

Time complexity. In Algorithm 1, the main costs include computing affinity matrices between features of different classes ($O(nd^2)$), constructing the graph Laplacian and filtered operators ($O(d^3)$), and performing spectral decompositions ($O(d^3)$). In Algorithm 2, this process is repeated N times during the iterative differential vector extraction phase. As a result, the total time complexity is $O\big(N(nd^2 + d^3)\big)$. This analysis reflects the computational cost associated with modeling the manifold of feature space and extracting class-specific latent variables. Note that when the feature dimension d is very large, the eigen-decomposition steps with complexity $O(d^3)$ may become a computational bottleneck. To address this, approximate methods such as randomized SVD or low-rank kernel approximations (e.g., Nyström method) can be employed in practice.
Space complexity. The dominant storage requirements come from kernel matrices, Laplacians, and filtered operators, all of size $d \times d$, as well as a small number of intermediate vectors. Therefore, the overall space complexity is $O(d^2)$.

D.2 Additional robustness analysis of PRISM

To further demonstrate the robustness of PRISM under small-sample scenarios, we evaluate its performance across varying train set sizes on the Madelon dataset. Figure 9 illustrates the relationship between the maximum classification accuracy and the number of training samples on a logarithmic scale. As shown, all methods except PRISM and ManiFeSt drop significantly as the sample size decreases, and they completely fail to identify relevant features when the training size falls below 10% (234 samples). In contrast, PRISM exhibits strong robustness to small sample size, maintaining competitive performance even when trained on only 1% of the data (23 samples).

D.3 Clusters on a hypercube

Due to the absence of ground-truth information on relevant features in the Madelon dataset, we construct a variant of Madelon [13] using the *make_classification* function from the *scikit-learn* library to enable more accurate evaluation of feature selection performance. In this variant, the ground-truth relevance of features is explicitly known, allowing for direct assessment of feature selection accuracy. More details on the dataset generation are in Appendix C.

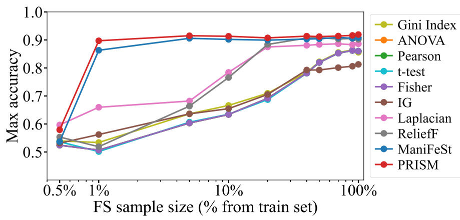

Figure 9: Max test accuracy vs. FS sample size on the Madelon dataset.

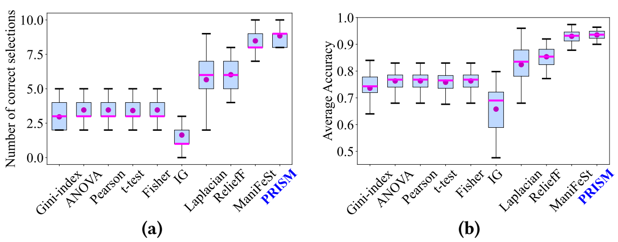

Figure 10: Results on the simulated variant of the Madelon dataset. (a) Boxplot of the number of correct selections. (b) Boxplot of the average accuracy.

The dataset consisting of 200 features is divided into train and test sets with 1500 and 500 samples, respectively. To show the effectiveness of PRISM under few-sample conditions, we perform feature selection using only 50 samples from the train set. For each method, the top 10 ranked features are selected, followed by training an SVM classifier optimized on the full train set. This procedure is repeated for 50 cross-validation iterations.

Figure 10a and 10b shows the number of correct selections and the average test accuracy achieved by the evaluated feature selection methods. We see that PRISM attains the highest number of correct selections (8.84) and the highest average test accuracy (0.9352) among all competing methods, surpassing the state-of-the-art approach ManiFeSt, which achieves 8.48 and 0.9301, respectively. These results clearly demonstrate the advantage of PRISM in few-sample scenarios.

D.4 More results on additional datasets

We compare PRISM with different FS methods on two additional datasets: Gisette and RELATHE. Dataset descriptions are in Appendix C. The results are shown in Table 5, including the colon cancer dataset. The numbers in parentheses indicate the number of features selected when each method achieved its best performance. As shown, PRISM consistently attains the highest classification accuracies across all datasets.

E Additional Theoretical Foundation

An l_2 norm convergence result for the random walk Laplacian $L^{(\mathrm{rw})} = I - D^{-1}K$ was derived in Cheng and Wu [8] under three assumptions: (i) the d observations are generated uniformly at random over a n-dimensional manifold $\mathcal{M}$, (ii) the smallest m eigenvalues

Table 5: Comparison of accuracy (%), standard deviation, and number of features used for various filter FS methods on benchmark datasets.

Method	Colon (2000/62)	Gisette (5000/7000)	RELATHE (4322/1427)
All features baseline	82.26 ± 13.69	97.96 ± 0.51	74.88 ± 3.85
Gini Index	83.43 ± 13.38 (20)	98.60 ± 0.50 (700)	82.94 ± 2.97 (100)
ANOVA	83.23 ± 12.58 (40)	98.65 ± 0.53 (800)	81.68 ± 3.31 (100)
Pearson	83.23 ± 12.58 (40)	98.65 ± 0.53 (800)	81.68 ± 3.31 (100)
t-test	82.32 ± 13.83 (600)	98.44 ± 0.47 (800)	83.26 ± 2.93 (100)
Fisher	83.23 ± 12.58 (40)	98.65 ± 0.53 (800)	81.68 ± 3.31 (100)
IG	85.05 ± 12.53 (40)	98.67 ± 0.52 (800)	81.19 ± 2.50 (90)
Laplacian	85.14 ± 12.51 (18)	98.60 ± 0.26 (1500)	75.94 ± 2.49 (100)
ReliefF	85.92 ± 12.86 (20)	98.67 ± 0.43 (1500)	82.03 ± 2.82 (100)
ManiFeSt	86.51 ± 12.72 (160)	98.27 ± 0.48 (1000)	77.48 ± 4.66 (100)
PRISM (ours)	**87.65** ± 13.91 (120)	**98.70** ± 0.45 (900)	**83.43** ± 2.11 (100)

of the LB operator over $\mathcal{M}$ have single multiplicity, with a minimal spectral gap $\eta_m > 0$, (iii) the bandwidth parameter of a Gaussian kernel satisfies $\sigma_d \to 0^+$ and $\sigma_d^{n/2+2} > C_m \frac{\log d}{d}$ for some constant C_m. Let $\phi_k(X) \in \mathbb{R}^d$ denote the vector of samples given by,

$$\phi_k(X) = \frac{1}{\sqrt{pd}} \beta_X(f_k), \tag{20}$$

where p denotes the uniform sampling density, f_k denotes the normalized eigenfunction corresponding to the k-th smallest eigenvalue of the LB operator, and $\beta_X(f_k) \in \mathbb{R}^d$ denotes the sampling operator that evaluates the eigenfunction f_k at a point set $X = \{x_1, \ldots, x_d\} \subseteq \mathcal{M}$. In addition, let $v_k \in \mathbb{R}^d$ and $v_k^{(\mathrm{rw})} \in \mathbb{R}^d$ denote the eigenvectors corresponding to the k-th smallest eigenvalues of the symmetric normalized Laplacian and the random walk Laplacian, respectively.

E.1 Theorem 2, Theorem 3 and Lemma 4

Theorem 2 (Theorem 5.5 [8]). *Under the assumptions (i)-(iii), for $d \to \infty$, with probability at least $1 - 4m^2 d^{-10} - (4m+6)d^{-9}$, the k-th eigenvector of the random walk Laplacian satisfies*

$$\left\| v_k^{(\mathrm{rw})} - \alpha\phi_k(X) \right\|_2 = O\left(\sigma_d + \sigma_d^{-n/4+1/2}\sqrt{\log d / d}\right), \quad \forall k \le m \tag{21}$$

where $v_k^{(\mathrm{rw})}$ is D-normalized such that $(v_k^{(\mathrm{rw})})^T D v_k^{(\mathrm{rw})}/(pd) = 1$, σ_d is the bandwidth parameter of the Gaussian kernel, and $|\alpha| = 1 + o_p(1)$.

Theorem 3. *Under the assumptions (i)-(iii), for $d \to \infty$, with probability at least $1 - 4m^2 d^{-10} - (4m+8)d^{-9}$, the k-th eigenvector of the symmetric normalized Laplacian satisfies*

$$\left\| v_k - \alpha\phi_k(X) \right\|_2 = O\left(\sigma_d + \sigma_d^{-n/4-1/2}\sqrt{\log d / d}\right), \quad \forall k \le m \tag{22}$$

where $\|v_k\| = 1$ is normalized and $|\alpha| = 1 + o_p(1)$.

The proof of Theorem 3 is in Appendix E.2.

Lemma 4. *Let $\mathcal{M}_1 = \mathcal{M}_a \times \mathcal{M}_s$ and $\mathcal{M}_2 = \mathcal{M}_b \times \mathcal{M}_s$. Let $v_{l,k}^{(1)}, v_{m,k'}^{(2)}$ denote the (l,k)-th and (m,k')-th unit-length eigenvectors of the symmetric normalized Laplacian matrices $L^{(1)}, L^{(2)}$, respectively. We assume that the corresponding eigenvalues $\mu_{l,k}^{(1)}$ and $\mu_{m,k'}^{(2)}$ are both within the m smallest eigenvalues of their respective spectra. Under the assumptions (i)-(iii), for $d \to \infty$, with probability at least $1 -$*

$12m^2d^{-10} - (8m+16)d^{-9}$, the inner product between $v_{l,k}^{(1)}$ and $v_{m,k'}^{(2)}$ satisfies

$$|(v_{l,k}^{(1)})^T v_{m,k'}^{(2)}| = \begin{cases} 1 - O\left(\sigma_d + \sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right), & \text{if } l = m = 0 \wedge k = k'. \\ O\left(\sigma_d + \sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right), & \text{otherwise.} \end{cases} \tag{23}$$

The proof of Lemma 4 is in Appendix E.5. The lemma plays a crucial role in deriving the convergence guarantee of Algorithm 1. It demonstrates that, apart from the eigenvectors associated only with the shared latent variables θ, the eigenvectors of the Laplacian matrices corresponding to two datasets are nearly orthogonal. Next, we aim to explain why the filtered operator in Eq. 8 is capable of removing the leading eigenvectors associated with the shared latent variables while retaining those associated only with the class-specific latent variables.

Let $M^{(1)} \in \mathbb{R}^{d\times m_1}$ denote the matrix whose columns consist of the eigenvectors of $L^{(1)}$ corresponding to the eigenvalues smaller than the threshold parameter τ of the filter $H(L^{(1)})$. We define a projection matrix as $P^{(1)} = I - M^{(1)}(M^{(1)})^T$, which is equal to the filter matrix $H(L^{(1)})$. For the analysis, we partition the columns of $M^{(1)}$ to two parts: (i) $M_\theta^{(1)}$ which contains only the eigenvectors associated with the shared latent variables θ, and (ii) $M_\varphi^{(1)}$ which contains the eigenvectors associated with the class-specific latent variables $\varphi^{(1)}$. Then, the following projection matrices can be defined:

$$P_\theta^{(1)} = I - M_\theta^{(1)}(M_\theta^{(1)})^T, \quad P_\varphi^{(1)} = I - M_\varphi^{(1)}(M_\varphi^{(1)})^T. \tag{24}$$

Due to the orthogonality of $M_\theta^{(1)}$ and $M_\varphi^{(1)}$, we have

$$P_\theta^{(1)}P_\varphi^{(1)} = I - M^{(1)}(M^{(1)})^T = P^{(1)}. \tag{25}$$

Similarly, let $M_\theta^{(2)}$ denote the matrix whose columns consists of the eigenvectors of $L^{(2)}$ associated with θ. And we denote a projection matrix by $P_\theta^{(2)} = I - M_\theta^{(2)}(M_\theta^{(2)})^T$. In step 7 of Algorithm 1, we apply a low pass filter to $Q^{(2)}$ and obtain $Q_\tau^{(2)}$ given by

$$Q_\tau^{(2)} = \sum_{l,k;\lambda_{l,k}^{(2)}\le\tau} (1-\lambda_{l,k}^{(2)}) v_{l,k}^{(2)}(v_{l,k}^{(2)})^T. \tag{26}$$

Thus, the filtered operator computed in step 7 of Algorithm 1 is equal to $\tilde{Q}^{(2)} = P^{(1)}Q_\tau^{(2)}P^{(1)}$. Note that the projection matrix $P_\theta^{(2)}$ is the *ideal filter* that perfectly eliminates the leading eigenvectors associated with the shared latent variables, whereas $P^{(1)}$ serves merely as a *practical filter*. Let $E_1 = P_\theta^{(2)}Q_\tau^{(2)}P_\theta^{(2)}$ and $E_2 = P_\varphi^{(1)}P_\theta^{(1)}Q_\tau^{(2)}P_\theta^{(1)}P_\varphi^{(1)} = \tilde{Q}^{(2)}$ denote the filtered operators obtained by applying the *ideal filter* and the *practical filter*, respectively. We bound the spectral norm of $E_1 - E_2$ in Appendix E.6.

E.2 Proof of Theorem 3

Let L be the symmetric normalized Laplacian and $L^{(\mathrm{rw})}$ the random walk Laplacian. The eigenvectors of L satisfy $v_k \propto D^{1/2}v_k^{(\mathrm{rw})}$, where D is the diagonal matrix of degrees and $v_k^{(\mathrm{rw})}$ is the corresponding eigenvector of $L^{(\mathrm{rw})}$ [42]. We take $v_k^{(\mathrm{rw})}$ D-normalized such that $(v_k^{(\mathrm{rw})})^T D v_k^{(\mathrm{rw})}/pd = 1$ as in Theorem 2 and $v_k = \frac{D^{1/2}}{\sqrt{pd}}v_k^{(\mathrm{rw})}$ so that $\|v_k\|^2 = \frac{(v_k^{(\mathrm{rw})})^T D v_k^{(\mathrm{rw})}}{pd} = 1$. By the triangle inequality, we have

$$\begin{aligned} \left\|v_k - \alpha\phi_k(X)\right\| &= \left\|\frac{D^{1/2}}{\sqrt{pd}}v_k^{(\mathrm{rw})} - \alpha\phi_k(X)\right\| \\ &= \left\|\frac{D^{1/2}}{\sqrt{pd}}v_k^{(\mathrm{rw})} - v_k^{(\mathrm{rw})} + v_k^{(\mathrm{rw})} - \alpha\phi_k(X)\right\| \\ &\le \left\|\frac{D^{1/2}}{\sqrt{pd}}v_k^{(\mathrm{rw})} - v_k^{(\mathrm{rw})}\right\| + \|v_k^{(\mathrm{rw})} - \alpha\phi_k(X)\|. \end{aligned} \tag{27}$$

Then we bound the two terms in the right-hand side separately. According to Theorem 2 [8, Theorem 5.5], the second term is bounded with probability at least $1 - 4m^2d^{-10} - (4m+6)d^{-9}$ as

$$\|v_k^{(\mathrm{rw})} - \alpha\phi_k(X)\| = O\left(\sigma_d + \sigma_d^{-n/4+1/2}\sqrt{\log d/d}\right). \tag{28}$$

To bound the first term, we need an additional bound on the degree matrix. According to the lemma [8, Lemma 3.5], for large values of d, with probability at least $1 - 2d^{-9}$, the following holds uniformly for all i:

$$D_{ii}/d = c_0 p + O\left(\sigma_d + \sigma_d^{-n/4}\sqrt{\log d/d}\right), \tag{29}$$

where c_0 is a constant determined by the choice of kernel and p is the uniform sampling density on the manifold. For the Gaussian kernel $c_0 = 1$, so we get

$$D_{ii}/pd = 1 + O\left(\sigma_d + \sigma_d^{-n/4}\sqrt{\log d/d}\right). \tag{30}$$

Then we take the square root of both sides and use a first order expansion of $\sqrt{1+x}$. Thus, we have

$$\sqrt{D_{ii}/pd} = 1 + O\left(\sigma_d + \sigma_d^{-n/4}\sqrt{\log d/d}\right). \tag{31}$$

Next, we bound the first term of the right-hand side of Eq. 27,

$$\left\|\frac{D^{1/2}}{\sqrt{pd}}v_k^{(\mathrm{rw})} - v_k^{(\mathrm{rw})}\right\| = \left\|\left(\frac{D^{1/2}}{\sqrt{pd}} - I\right)v_k^{(\mathrm{rw})}\right\| \le \left\|\left(\frac{D^{1/2}}{\sqrt{pd}} - I\right)\right\|\|v_k^{(\mathrm{rw})}\|, \tag{32}$$

where the first term in the right-hand side of Eq. 32 denotes the operator norm. We know that the operator norm of a diagonal matrix is the maximum absolute value of the diagonal elements, given by

$$\left\|\left(\frac{D^{1/2}}{\sqrt{pd}} - I\right)\right\| = \max_i \left|\frac{D_{ii}^{1/2}}{\sqrt{pd}} - 1\right| = O\left(\sigma_d + \sigma_d^{-n/4}\sqrt{\log d/d}\right), \tag{33}$$

where this bound holds uniformly for all i with probability at least $1 - 2d^{-9}$. Then we derive a bound for $\|v_k^{(\mathrm{rw})}\|$. Note that $v_k^{(\mathrm{rw})}$ is D-normalized such that $(v_k^{(\mathrm{rw})})^T D v_k^{(\mathrm{rw})}/pd = 1$, hence

$$\frac{pd}{\max_i D_{ii}} \le \|v_k^{(\mathrm{rw})}\|^2 \le \frac{pd}{\min_i D_{ii}}. \tag{34}$$

By combining Eq. 30 with the first order expansion of $1/(1+x)$, we obtain that both the lower and upper bounds in Eq. 34 are

$1 + O\left(\sigma_d + \sigma_d^{-n/4}\sqrt{\log d/d}\right)$. Thus, by a first order expansion of $\sqrt{1+x}$, we have

$$\|v_k^{(\mathrm{rw})}\| = \sqrt{\|v_k^{(\mathrm{rw})}\|^2} = 1 + O\left(\sigma_d + \sigma_d^{-n/4}\sqrt{\log d/d}\right). \tag{35}$$

Plugging Eq. 35 and Eq. 33 back into Eq. 32 yields

$$\left\|\frac{D^{1/2}}{\sqrt{pd}}v_k^{(\mathrm{rw})} - v_k^{(\mathrm{rw})}\right\| = O\left(\sigma_d + \sigma_d^{-n/4}\sqrt{\log d/d}\right). \tag{36}$$

Finally, plugging Eq. 28 and Eq. 36 back into Eq. 27 and applying the union bound over the events where either bound may fail, we conclude that, with probability at least

$$1 - [4m^2d^{-10} + (4m+6)d^{-9}] - 2d^{-9} = 1 - 4m^2d^{-10} - (4m+8)d^{-9}, \tag{37}$$

the following bound holds

$$\|v_k - \alpha\phi_k(X)\| = O\left(\sigma_d + \sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right) \qquad \forall k \le m. \tag{38}$$

E.3 Auxiliary lemmas for the proof of Lemma 4 and Theorem 1

Lemma 5. *Let $v^{(1)} = u^{(1)} - \sigma^{(1)}$ and $v^{(2)} = u^{(2)} - \sigma^{(2)}$ be two vectors that satisfy:*

(1) $\|v^{(1)}\| = \|v^{(2)}\| = 1$.
(2) $\|\sigma^{(1)}\|, \|\sigma^{(2)}\| = O(\sigma_d)$.
(3) The vector $u^{(1)}$ is proportional to $u^{(2)}$ such that $u^{(1)} = cu^{(2)}$ for some constant c.

Then $|(v^{(1)})^Tv^{(2)}| = 1 - O(\sigma_d)$.

Lemma 6. *Let $U, V \in \mathbb{R}^{d\times m}$ be orthogonal matrices with columns u_i and v_i, respectively. Assume that for some $\varepsilon > 0$,*

$$u_i^Tv_i \ge 1 - \varepsilon \qquad \forall i = 1, \dots, m. \tag{39}$$

Then $\|U - V\|^2 \le 2m\varepsilon$.

Lemma 7. *Let M be a symmetric positive semi-definite matrix. Let $U, V \in \mathbb{R}^{d\times m}$ be two orthogonal matrices such that $U^TU = V^TV = I$, and let $P_1 = I - UU^T$ and $P_2 = I - VV^T$ be two projection matrices. Then,*

$$\|P_1MP_1 - P_2MP_2\| \le 4\|M\|\|U - V\|. \tag{40}$$

Lemma 8. *Let $A \in \mathbb{R}^{d\times d}$ be a symmetric positive semi-definite matrix with spectral decomposition $A = \sum_k \lambda_k u_k u_k^T$, where u_k, λ_k are eigenvectors and eigenvalues respectively. Let $V \in \mathbb{R}^{d\times m}$ be a matrix whose columns $\{v_i\}_{i=1}^m$ are orthogonal. Suppose that $|v_i^Tu_j| \le \sigma$ holds for all (i, j), where $\sigma \ge 1/\sqrt{d}$. Then*

$$\|A - (I - VV^T)A(I - VV^T)\| \le m\sigma^2\sum_{k=1}^d \lambda_k. \tag{41}$$

E.4 Proofs of auxiliary lemmas

Proof of Lemma 5. We first use the reverse triangle inequality to derive a bound over $u^{(1)}$ and $u^{(2)}$,

$$||u^{(1)}|| - ||\sigma^{(1)}|| \le ||v^{(1)}|| = ||u^{(1)} - \sigma^{(1)}|| \le ||u^{(1)}|| + ||\sigma^{(1)}||. \tag{42}$$

Thus, combined with assumptions (1) and (2), we get

$$1 + O(\sigma_d) \ge ||u^{(1)}|| \ge 1 - O(\sigma_d). \tag{43}$$

A similar bound can be derived for $||u^{(2)}||$. Next, we derive a bound for $|(u^{(1)})^Tu^{(2)}|$. Since $u^{(1)} = cu^{(2)}$, then $\frac{|(u^{(1)})^Tu^{(2)}|}{||u^{(1)}||||u^{(2)}||} = 1$. Thus, combining Eq. 43 and the above, we have

$$|(u^{(1)})^Tu^{(2)}| = ||u^{(1)}|| \cdot ||u^{(2)}|| \ge 1 - O(\sigma_d). \tag{44}$$

Finally, we derive a bound for $|(v^{(1)})^Tv^{(2)}|$. By the reverse triangle inequality, we have

$$\begin{aligned}|(v^{(1)})^Tv^{(2)}| &= |(u^{(1)} - \sigma^{(1)})^T(u^{(2)} - \sigma^{(2)})|\\ &\ge |(u^{(1)})^Tu^{(2)}| - |(\sigma^{(1)})^Tu^{(2)}|\\ &\quad - |(\sigma^{(2)})^Tu^{(1)}| - |(\sigma^{(1)})^T\sigma^{(2)}|.\end{aligned} \tag{45}$$

According to Eq. 44, the first term is lower bounded by $1 - O(\sigma_d)$. By the assumption (2), the fourth term is bounded by $O(\sigma_d^2)$. Using Eq. 43, the second and third terms can be bounded by Cauchy-Schwarz inequality as follows,

$$|(\sigma^{(1)})^Tu^{(2)}| \le ||\sigma^{(1)}|| \cdot ||u^{(2)}|| = O(\sigma_d). \tag{46}$$

Thus, we demonstrate that

$$|(v^{(1)})^Tv^{(2)}| \ge 1 - O(\sigma_d). \tag{47}$$

Proof of Lemma 6. Since $u_i^Tv_i \ge 1 - \varepsilon$, we have

$$\|u_i - v_i\|^2 = \|u_i\|^2 + \|v_i\|^2 - 2u_i^Tv_i = 2(1 - u_i^Tv_i) \le 2\varepsilon. \tag{48}$$

The spectral norm of a matrix is bounded by its Frobenius norm. Thus,

$$\|U - V\|^2 \le \|U - V\|_F^2 = \sum_{i=1}^m \|u_i - v_i\|^2 \le 2m\varepsilon. \tag{49}$$

Proof of Lemma 7. Since M is symmetric, $P_1MP_1 - P_2MP_2$ is also symmetric. Thus the spectral norm is equal to the largest eigenvalue, given by

$$\|P_1MP_1 - P_2MP_2\| = \max_{\|x\|=1}\left|x^T(P_1MP_1 - P_2MP_2)x\right|. \tag{50}$$

As M is positive semi-definite, it has a square root, denoted by $M^{1/2}$. Thus, for any vector x we have

$$\begin{aligned}&|x^T(P_1MP_1 - P_2MP_2)x|\\ &= |x^TP_1MP_1x - x^TP_2MP_2x|\\ &= \left|\|M^{1/2}P_1x\|^2 - \|M^{1/2}P_2x\|^2\right|\\ &= \left|\|M^{1/2}P_1x\| + \|M^{1/2}P_2x\|\right| \cdot \left|\|M^{1/2}P_1x\| - \|M^{1/2}P_2x\|\right|.\end{aligned} \tag{51}$$

By Cauchy-Schwarz inequality we have $\|M^{1/2}P_1x\| \le \|M^{1/2}\|$ and $\|M^{1/2}P_2x\| \le \|M^{1/2}\|$. Thus,

$$\left|\|M^{1/2}P_1x\| + \|M^{1/2}P_2x\|\right| \le 2\|M^{1/2}\|. \tag{52}$$

By the reverse triangle inequality, we have

$$\begin{aligned}\left|\|M^{1/2}P_1x\| - \|M^{1/2}P_2x\|\right| &\le \left\|M^{1/2}(P_1 - P_2)x\right\|\\ &\le \|M^{1/2}\|\|P_1 - P_2\|\\ &= \|M^{1/2}\|\|UU^T - VV^T\|.\end{aligned} \tag{53}$$

We apply the reverse triangle inequality again to bound the norm $\|UU^T - VV^T\|$. For any vector x we have

$$\begin{aligned}\left|x^T(UU^T - VV^T)x\right| &= \left|x^T UU^T x - x^T VV^T x\right| = \left|\|U^T x\|^2 - \|V^T x\|^2\right| \\ &= \left|\|U^T x\| + \|V^T x\|\right| \cdot \left|\|U^T x\| - \|V^T x\|\right| \\ &\leq 2\left\|(U-V)^T x\right\| \leq 2\|U - V\|. \end{aligned} \tag{54}$$

Thus,

$$\|UU^T - VV^T\| = \max_{\|x\|=1} \left|x^T(UU^T - VV^T)x\right| \leq 2\|U - V\|. \tag{55}$$

Combining the bounds in Eqs. 51, 52, 53 and 55 yields

$$\|P_1 M P_1 - P_2 M P_2\| \leq 2\|M\| \cdot \|UU^T - VV^T\| \leq 4\|M\|\|U - V\|. \tag{56}$$

Proof of Lemma 8. Since A is symmetric, $A - (I - VV^T)A(I - VV^T)$ is also symmetric. Thus the spectral norm is equal to the largest eigenvalue, given by

$$\|A - (I - VV^T)A(I - VV^T)\| = \max_{\|z\|=1} \left|z^T\left(A - (I - VV^T)A(I - VV^T)\right)z\right|. \tag{57}$$

First, we prove that the maximizer z^* of Eq. 57 is a linear combination of the vectors in $\{v_i\}_{i=1}^m$ such that $z^* = \sum_{j=1}^m \alpha_j v_j$. Any vector z orthogonal to V satisfies,

$$z^T\left(A - (I - VV^T)A(I - VV^T)\right)z = z^T A z - z^T A z = 0. \tag{58}$$

Thus, the matrix $\left(A - (I - VV^T)A(I - VV^T)\right)$ has a zero eigenvalue with multiplicity $d - m$. Since it is a symmetric matrix, it has exactly m eigenvectors corresponding to non-zero eigenvalues, and these eigenvectors are contained within the span of $\{v_i\}_{i=1}^m$. This implies that the leading eigenvector z^* is a linear combination of $\{v_i\}_{i=1}^m$. Thus, we have $VV^T z^* = z^*$, and hence

$$(z^*)^T\left(A - (I - VV^T)A(I - VV^T)\right)z^* = (z^*)^T A z^*. \tag{59}$$

Next, we derive a bound over $|v_j^T A v_i|$ for every pair of vectors v_i, v_j. Since $|v_i^T u_j| \leq \sigma$, we have

$$|v_j^T A v_i| = \left|v_j^T \sum_{k=1}^d \lambda_k u_k u_k^T v_i\right| \leq \sum_{k=1}^d \lambda_k |v_j^T u_k||v_i^T u_k| \leq \sigma^2 \sum_{k=1}^d \lambda_k. \tag{60}$$

Let $z^* = \sum_{i=1}^m \alpha_i v_i$. By Eqs. 59 and 60, we show that the maximal eigenvalue is bounded by

$$|(z^*)^T A z^*| = \left|\sum_{ij} \alpha_i \alpha_j v_i^T A v_j\right| \leq \sum_{ij} |\alpha_i \alpha_j v_i^T A v_j| \leq \sum_{k=1}^d \lambda_k \sigma^2 \sum_{ij} |\alpha_i \alpha_j|. \tag{61}$$

Finally, by Cauchy-Schwarz inequality we have that $\sum_{ij} |\alpha_i \alpha_j| = \left(\sum_i |\alpha_i|\right)^2 \leq m \sum_i \alpha_i^2$. Under the unit norm constraint $\sum_i \alpha_i^2 = 1$, the maximum value of $\sum_i |\alpha_i|$ is obtained for $\alpha_i = 1/\sqrt{m}$. Thus, $\sum_{ij} |\alpha_i \alpha_j| \leq m$. Combining Eq. 61 and the above completes the proof.

E.5 Proof of Lemma 4

By Theorem 3, for all eigenfunctions $f_{l,k}^{(1)}$ corresponding to eigenvalues $\mu_{l,k}^{(1)}$ that are among the smallest m eigenvalues of the Laplace-Beltrami operator on $\mathcal{M}^{(1)}$, with probability at least $1 - 4m^2 d^{-10} - (4m+8)d^{-9}$ there exists a constant $\alpha^{(1)}$ such that the unit-normalized eigenvector $v_{l,k}^{(1)}$ satisfies

$$\left\|\frac{\alpha^{(1)}}{\sqrt{p^{(1)} d}} \beta_X(f_{l,k}^{(1)}) - v_{l,k}^{(1)}\right\|_2 = O\left(\sigma_d + \sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right). \tag{62}$$

The same result holds for $\alpha^{(2)}, f_{m,k'}^{(2)}, \mu_{m,k'}^{(2)}$ and $v_{m,k'}^{(2)}$. We denote the differences by

$$\sigma_{l,k}^{(1)} = \frac{\alpha^{(1)}}{\sqrt{p^{(1)} d}} \beta_X(f_{l,k}^{(1)}) - v_{l,k}^{(1)}, \quad \sigma_{m,k'}^{(2)} = \frac{\alpha^{(2)}}{\sqrt{p^{(2)} d}} \beta_X(f_{m,k'}^{(2)}) - v_{m,k'}^{(2)}. \tag{63}$$

To bound the inner product of $v_{l,k}^{(1)}$ and $v_{m,k'}^{(2)}$ we apply the triangle inequality,

$$\begin{aligned} &|(v_{l,k}^{(1)})^T v_{m,k'}^{(2)}| \\ &= \left|\left(\frac{\alpha^{(1)}}{\sqrt{p^{(1)} d}} \beta_X(f_{l,k}^{(1)}) - \sigma_{l,k}^{(1)}\right)^T \left(\frac{\alpha^{(2)}}{\sqrt{p^{(2)} d}} \beta_X(f_{m,k'}^{(2)}) - \sigma_{m,k'}^{(2)}\right)\right| \\ &\leq \frac{\alpha^{(1)}\alpha^{(2)}}{d\sqrt{p^{(1)} p^{(2)}}} |\beta_X(f_{l,k}^{(1)})^T \beta_X(f_{m,k'}^{(2)})| + \frac{\alpha^{(2)}}{\sqrt{p^{(2)} d}} |(\sigma_{l,k}^{(1)})^T \beta_X(f_{m,k'}^{(2)})| \\ &\quad + \frac{\alpha^{(1)}}{\sqrt{p^{(1)} d}} |(\beta_X(f_{l,k}^{(1)})^T \sigma_{m,k'}^{(2)}| + |(\sigma_{l,k}^{(1)})^T \sigma_{m,k'}^{(2)}|. \end{aligned} \tag{64}$$

Let us address each of these terms separately. The fourth term of Eq. 64 is bounded by

$$|(\sigma_{l,k}^{(1)})^T \sigma_{m,k'}^{(2)}| \leq \|\sigma_{l,k}^{(1)}\|\|\sigma_{m,k'}^{(2)}\| = O\left(\left(\sigma_d + \sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right)^2\right). \tag{65}$$

Thus, it is negligible. The second and third terms in Eq. 64 can be bounded via the Cauchy-Schwarz inequality. For example, the second term is bounded by

$$\frac{\alpha^{(2)}}{\sqrt{p^{(2)} d}} |(\sigma_{l,k}^{(1)})^T \beta_X(f_{m,k'}^{(2)})| \leq \frac{\alpha^{(2)}}{\sqrt{p^{(2)} d}} \|\sigma_{l,k}^{(1)}\|\|\beta_X(f_{m,k'}^{(2)})\|. \tag{66}$$

By Lemma 3.4 in Cheng and Wu [8], with probability at least $1 - 2m^2 d^{-10}$, the term $\frac{1}{p^{(2)} d}\|\beta_X(f_{m,k'}^{(2)})\|^2$ is $1 + O_p(\log d/d)$. Thus, by a first order expansion of $\sqrt{1+x}$, we have

$$\frac{1}{\sqrt{p^{(2)} d}} \|\beta_X(f_{m,k'}^{(2)})\| = 1 + O_p(\log d/d). \tag{67}$$

Combining this result with the bounds on $\|\sigma_{l,k}^{(1)}\|, \|\sigma_{m,k'}^{(2)}\|$ in Theorem 3, we have

$$\begin{aligned} &\frac{\alpha^{(2)}}{\sqrt{p^{(2)} d}} |(\sigma_{l,k}^{(1)})^T \beta_X(f_{m,k'}^{(2)})| + \frac{\alpha^{(1)}}{\sqrt{p^{(1)} d}} |(\beta_X(f_{l,k}^{(1)})^T \sigma_{m,k'}^{(2)}| \\ &= O\left(\sigma_d + \sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right). \end{aligned} \tag{68}$$

We now bound the first term of Eq. 64. It is equal to,

$$\begin{aligned}
&\frac{\alpha^{(1)}\alpha^{(2)}}{d\sqrt{p^{(1)}p^{(2)}}}\left|\beta_X(f^{(1)}_{l,k})^T\beta_X(f^{(2)}_{m,k'})\right| \\
&= \frac{\alpha^{(1)}\alpha^{(2)}}{d\sqrt{p^{(1)}p^{(2)}}}\Bigg|\left(\beta_{\pi_a(X)}(f^{(a)}_l)\cdot\beta_{\pi_s(X)}(f^{(s)}_k)\right)^T \\
&\quad\times\left(\beta_{\pi_b(X)}(f^{(b)}_m)\cdot\beta_{\pi_s(X)}(f^{(s)}_{k'})\right)\Bigg| \\
&= \frac{\alpha^{(1)}\alpha^{(2)}}{d\sqrt{p^{(1)}p^{(2)}}}\Bigg|\sum_{i=1}^{d} f^{(a)}_l(\pi_a(x_i))f^{(b)}_m(\pi_b(x_i)) \\
&\quad\cdot f^{(s)}_k(\pi_s(x_i))f^{(s)}_{k'}(\pi_s(x_i))\Bigg|. \qquad (69)
\end{aligned}$$

Consider the summands in Eq. 69. Since the coordinates on different manifolds are sampled independently in our setting, the expectation of the summands factorizes,

$$\begin{aligned}
&\mathbb{E}[f^{(a)}_l(\pi_a(x))f^{(b)}_m(\pi_b(x))f^{(s)}_k(\pi_s(x))f^{(s)}_{k'}(\pi_s(x))] \\
&= \mathbb{E}[f^{(a)}_l(\pi_a(x))]\mathbb{E}[f^{(b)}_m(\pi_b(x))]\mathbb{E}[f^{(s)}_k(\pi_s(x))f^{(s)}_{k'}(\pi_s(x))]. \qquad (70)
\end{aligned}$$

By the orthogonality of eigenfunctions with different eigenvalues, we have

$$\begin{aligned}
&\mathbb{E}[f^{(a)}_a(\pi_a(x))] = 0 \quad \forall l > 0, \\
&\mathbb{E}[f^{(b)}_m(\pi_b(x))] = 0 \quad \forall m > 0, \qquad (71)\\
&\mathbb{E}[f^{(s)}_k(\pi_s(x))f^{(s)}_{k'}(\pi_s(x))] = 0 \quad \forall (k \neq k').
\end{aligned}$$

Hence, except in the special case $l = m = 0$ and $k = k'$, each summand is a mean-zero random variable. Moreover, by Cauchy-Schwarz inequality we can bound the second moment,

$$\begin{aligned}
&\mathbb{E}[(f^{(a)}_l(\pi_a(x)))^2] = 1 \quad \forall l, \\
&\mathbb{E}[(f^{(b)}_m(\pi_b(x)))^2] = 1 \quad \forall m, \qquad (72)\\
&\mathbb{E}\left[\left(f^{(s)}_k(\pi_s(x))f^{(s)}_{k'}(\pi_s(x))\right)^2\right] \leq 1 \quad \forall (k, k').
\end{aligned}$$

By Eqs. 70, 71 and 72, we obtain that unless $l = m = 0$ and $k = k'$, the sum in Eq. 69 is over i.i.d random variables that are centred with variance ≤ 1. Since the terms are i.i.d, the variance of the sum is bounded by d. Thus by Chebyshev's inequality, the sum is $O_p(\sqrt{d})$. Therefore,

$$\frac{\alpha^{(1)}\alpha^{(2)}}{d\sqrt{p^{(1)}p^{(2)}}}|\beta_X(f^{(1)}_{l,k})^T\beta_X(f^{(2)}_{m,k'})| = \frac{\alpha^{(1)}\alpha^{(2)}}{\sqrt{p^{(1)}p^{(2)}}}O_p(1/\sqrt{d}) = O_p(1/\sqrt{d}). \qquad (73)$$

Then we compare the relative magnitudes of σ_d and $1/\sqrt{d}$. Note that if $\sigma_d \geq 1$ then $1/\sqrt{d} < \sigma_d$ and if $\sigma_d < 1$ then $1/\sqrt{d} < \sigma_d^{-n/4-1/2}\sqrt{\log d/d}$. In both cases, $1/\sqrt{d}$ is negligible compared with the other terms given in Eqs. 65 and 68. Thus for $k \neq k'$, combining the bounds on the four terms and applying the union bound over the events where either bound may fail, we conclude that, with probability at least

$$\begin{aligned}
&1 - \left[2\left(4m^2d^{-10} + (4m+8)d^{-9}\right) + 2\left(2m^2d^{-10}\right)\right] \\
&= 1 - 12m^2d^{-10} - (8m+16)d^{-9}, \qquad (74)
\end{aligned}$$

the following bound holds

$$(v^{(1)}_{l,k})^T v^{(2)}_{m,k'} = O\left(\sigma_d + \sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right). \qquad (75)$$

The only remaining case to consider is when $l = m = 0$ and $k = k'$. Since $f^{(a)}_0, f^{(b)}_0$ are both constant functions, then by Eq. 16 we have

$$\beta_X(f^{(1)}_{0,k}) \sim \beta_{\pi_a(X)}(f^{(a)}_0)\beta_{\pi_s(X)}(f^{(s)}_k) \sim \beta_{\pi_s(X)}f^{(s)}_k. \qquad (76)$$

A similar derivation holds for $\beta_X(f^{(2)}_{0,k'})$. Therefore, we show that $\beta_X(f^{(1)}_{0,k}) \sim \beta_X(f^{(2)}_{0,k'})$. Let $u^{(1)} = \alpha^{(1)}/\sqrt{p^{(1)}d}\cdot\beta_X(f^{(1)}_{0,k})$ and $u^{(2)} = \alpha^{(2)}/\sqrt{p^{(2)}d}\cdot\beta_X(f^{(2)}_{0,k})$. Applying Lemma 5 to $u^{(1)}, u^{(2)}$ together with $\sigma^{(1)}_{0,k}, \sigma^{(2)}_{0,k}$ yields

$$|(v^{(1)}_{0,k})^T v^{(2)}_{0,k}| = 1 - O(\|\sigma^{(1)}_{0,k}\| + \|\sigma^{(2)}_{0,k}\|). \qquad (77)$$

Combining this result with the bounds on $\|\sigma^{(1)}_{0,k}\|, \|\sigma^{(2)}_{0,k}\|$ given in Theorem 3, we obtain that, if $l = m = 0$ and $k = k'$, then

$$|(v^{(1)}_{0,k})^T v^{(2)}_{0,k}| = 1 - O\left(\sigma_d + \sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right). \qquad (78)$$

This completes the proof.

E.6 Proof of Theorem 1

Note that $M^{(2)}_\theta$ contains the eigenvectors of $Q^{(2)}_\tau$ associated with the shared latent variables θ. Thus, projecting $Q^{(2)}_\tau$ onto the orthogonal complement of its leading eigenvectors $\{v^{(2)}_{0,1}, v^{(2)}_{0,2}, \ldots, v^{(2)}_{0,m}\}$ gives the matrix $E_1 = P^{(2)}_\theta Q^{(2)}_\tau P^{(2)}_\theta$, which eliminates those eigenvectors. Due to the eigenvalue structure of the product manifold $\mathcal{M}_2$ (see Eq. 17), we obtain that for a sufficiently large m, the leading eigenvector of E_1 is $v^{(2)}_{1,0}$, which is the leading eigenvector that is not associated with θ. Thus, we have

$$\underset{\|v\|=1}{\arg\max}\ v^T E_1 v = v^{(2)}_{1,0}. \qquad (79)$$

Theorem 3 then implies that, as $d \to \infty$,

$$\left\|v^{(2)}_{1,0} - \frac{\alpha}{\sqrt{pd}}\beta_X(f^{(b)}_{1,0})\right\| = O\left(\sigma_d + \sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right). \qquad (80)$$

To control the deviation between $\delta^{(2)}$ and $v^{(2)}_{1,0}$, it suffices to bound the spectral norm $\|E_2 - E_1\|$, where $E_2 = P^{(1)}_\varphi P^{(1)}_\theta Q^{(2)}_\tau P^{(1)}_\theta P^{(1)}_\varphi$. Adding and subtracting $P^{(1)}_\varphi P^{(2)}_\theta Q^{(2)}_\tau P^{(2)}_\theta P^{(1)}_\varphi$ and then applying the triangle inequality yields

$$\begin{aligned}
\|E_2 - E_1\| &= \|P^{(1)}_\varphi P^{(1)}_\theta Q^{(2)}_\tau P^{(1)}_\theta P^{(1)}_\varphi - P^{(2)}_\theta Q^{(2)}_\tau P^{(2)}_\theta\| \\
&\leq \|P^{(1)}_\varphi P^{(1)}_\theta Q^{(2)}_\tau P^{(1)}_\theta P^{(1)}_\varphi - P^{(1)}_\varphi P^{(2)}_\theta Q^{(2)}_\tau P^{(2)}_\theta P^{(1)}_\varphi\| \\
&\quad + \|P^{(1)}_\varphi P^{(2)}_\theta Q^{(2)}_\tau P^{(2)}_\theta P^{(1)}_\varphi - P^{(2)}_\theta Q^{(2)}_\tau P^{(2)}_\theta\|. \qquad (81)
\end{aligned}$$

We now bound the first term of the right-hand side of Eq. 81,

$$\begin{aligned}
&\|P^{(1)}_\varphi P^{(1)}_\theta Q^{(2)}_\tau P^{(1)}_\theta P^{(1)}_\varphi - P^{(1)}_\varphi P^{(2)}_\theta Q^{(2)}_\tau P^{(2)}_\theta P^{(1)}_\varphi\| \\
&= \|P^{(1)}_\varphi (P^{(1)}_\theta Q^{(2)}_\tau P^{(1)}_\theta - P^{(2)}_\theta Q^{(2)}_\tau P^{(2)}_\theta) P^{(1)}_\varphi\| \\
&\leq \|P^{(1)}_\theta Q^{(2)}_\tau P^{(1)}_\theta - P^{(2)}_\theta Q^{(2)}_\tau P^{(2)}_\theta\| \\
&\leq 4\|Q^{(2)}_\tau\|\|M^{(1)}_\theta - M^{(2)}_\theta\|. \qquad (82)
\end{aligned}$$

Since $P^{(1)}_\varphi$ is a projection matrix, then $\|P^{(1)}_\varphi\| \leq 1$. By the inequality $\|ABC\| \leq \|A\|\|B\|\|C\|$ that holds for all operator norms, we can prove the first inequality of Eq. 82. The second inequality of Eq. 82 is proved in Lemma 7. Then, combining Lemma 4 with Lemma 6, we get

$$\|M_\theta^{(2)} - M_\theta^{(1)}\|^2 = O_p\left(m\sigma_d + m\sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right). \tag{83}$$

Thus, combining this result with the bound on the eigenvalues of the operator, $\|Q^{(2)}\| \leq 1$, yields

$$\begin{aligned}
&\|P_\varphi^{(1)} P_\theta^{(1)} Q_\tau^{(2)} P_\theta^{(1)} P_\varphi^{(1)} - P_\varphi^{(1)} P_\theta^{(2)} Q_\tau^{(2)} P_\theta^{(2)} P_\varphi^{(1)}\|^2 \\
&\leq 16\|Q_\tau^{(2)}\|^2 \|M_\theta^{(1)} - M_\theta^{(2)}\|^2 \\
&\leq O(m\sigma_d) + O\left(m\sqrt{\frac{\log d}{d\sigma_d^{n/2+1}}}\right).
\end{aligned} \tag{84}$$

Since each of the bounds above holds with probability at least $1 - 2m^2 d^{-10}$ or $1 - (2m+6)d^{-9}$, applying the union bound over all events yields that, all the inequalities above hold simultaneously with probability at least

$$1 - \left[2(2m^2 d^{-10}) + (2m+6)d^{-9}\right] = 1 - 4m^2 d^{-10} - (2m+6)d^{-9}. \tag{85}$$

Next, we bound the second term of the right-hand side of Eq. 81. Applying Lemma 8 with the matrix $P_\theta^{(2)} Q_\tau^{(2)} P_\theta^{(2)}$ and projection matrix $P_\varphi^{(1)}$, we get

$$\|P_\varphi^{(1)} P_\theta^{(2)} Q_\tau^{(2)} P_\theta^{(2)} P_\varphi^{(1)} - P_\theta^{(2)} Q_\tau^{(2)} P_\theta^{(2)}\| \leq m\sigma^2 \Big(\sum_{l,k;\lambda_{l,k}^{(1)} < \tau} (1 - \lambda_{l,k}^{(1)}) \Big). \tag{86}$$

The bound σ on the inner products between the eigenvectors of $Q_\tau^{(2)}$ and a vector in the span of $P_\theta^{(2)}$, as required in Lemma 8 is given by Lemma 4. Thus, we have

$$\sigma = O\left(\sigma_d + \sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right). \tag{87}$$

Plugging this back into Eq. 86 yields

$$\begin{aligned}
&\|P_\varphi^{(1)} P_\theta^{(2)} Q_\tau^{(2)} P_\theta^{(2)} P_\varphi^{(1)} - P_\theta^{(2)} Q_\tau^{(2)} P_\theta^{(2)}\| \\
&= \Big(\sum_{l,k;\lambda_{l,k}^{(1)} < \tau} (1 - \lambda_{l,k}^{(1)}) \Big) O\left(m\sigma_d + m\sqrt{\frac{\log d}{d\sigma_d^{n/2+1}}}\right) \\
&= O(m^2\sigma_d) + O\left(m^2\sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right).
\end{aligned} \tag{88}$$

Since the convergence rate in Eq. 84 is slower than Eq. 88, Then the overall convergence rate of $\|E_2 - E_1\|^2$ is

$$\|E_2 - E_1\|^2 = O(m\sigma_d) + O\left(m\sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right). \tag{89}$$

At this point, the Davis-Kahan theorem can be applied to the symmetric matrices E_1 and E_2 and their respective leading eigenvectors $v_{1,0}^{(2)}$ and $\delta^{(2)}$. According to the theorem, given that $(\delta^{(2)})^T v_{1,0}^{(2)} \geq 0$, we have [45, Corollary 1]

$$\|\delta^{(2)} - v_{1,0}^{(2)}\|^2 \leq \frac{2^{\frac{4}{3}}\|E_2 - E_1\|^2}{\eta_m^2}, \tag{90}$$

where η_m is the minimal spectral gap, which is larger than zero under the assumption (ii). Thus, combining Eq. 89 with Eq. 90 yields

$$\|\delta^{(2)} - v_{1,0}^{(2)}\|^2 \leq O(m\sigma_d) + O\left(m\sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right). \tag{91}$$

By the triangle inequality, we have

$$\begin{aligned}
&\|\delta^{(2)} - \frac{\alpha}{\sqrt{pd}}\beta_{\pi_b(x)}(f_1^{(b)})\|^2 \\
&= \|\delta^{(2)} - v_{1,0}^{(2)} + v_{1,0}^{(2)} - \frac{\alpha}{\sqrt{pd}}\beta_X(f_{1,0}^{(b)})\|^2 \\
&\leq 2\|\delta^{(2)} - v_{1,0}^{(2)}\|^2 + 2\|v_{1,0}^{(2)} - \frac{\alpha}{\sqrt{pd}}\beta_X(f_{1,0}^{(b)})\|^2 \\
&= O(\|\delta^{(2)} - v_{1,0}^{(2)}\|^2) + O(\|v_{1,0}^{(2)} - \frac{\alpha}{\sqrt{pd}}\beta_X(f_{1,0}^{(b)})\|^2).
\end{aligned} \tag{92}$$

Finally, substituting Eq. 80 and Eq. 91 into Eq. 92 and applying the union bound over the events where either bound may fail, we conclude that, with probability at least $1 - 4m^2 d^{-10} - (2m+6)d^{-9}$, the following bound holds

$$\|\delta^{(2)} - \frac{\alpha}{\sqrt{pd}}\beta_{\pi_b(x)}(f_1^{(b)})\|^2 \leq O(m\sigma_d) + O\left(m\sigma_d^{-n/4-1/2}\sqrt{\log d/d}\right). \tag{93}$$

This completes the proof.